EVALUATING EVAPORATION FROM LAND SURFACES NEAR THE GREAT SALT LAKE

by

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ABSTRACT

The desiccation of saline lakes, such as the Great Salt Lake (GSL), presents significant environmental challenges, including increased water insecurity and habitat destruction. This phenomenon is partially dependent on natural variability and climate change, but is largely influenced by excessive water use and management policy. An essential component of water balance within closed lake systems is the amount that is lost to evaporation and evapotranspiration. These processes within the GSL basin have been incompletely studied.

This study evaluates evaporation fluxes from both exposed playa and anthropogenic mineral pond surfaces using two eddy covariance sensor systems: the open-path, three-dimensional Campbell Scientific EC150 and the semi-open-path, one-dimensional LI-COR LI-710. LI-COR's release of the lower cost LI-710 during 2024 provided the impetus for comparing its performance characteristics to those of the EC150. Analyses are conducted of daily evaporation rates, energy flux patterns, and sensor performance under distinct environmental conditions. Instruments were deployed at a playa site in Farmington Bay and at a mineral pond site operated by Compass Minerals, with data evaluated in this study from October 2024 through March 2025. The procedures used to assess sensor performance will be discussed in detail, using data collected at these sites

Differences in flux rates between the playa and mineral pond sites highlight the impact of surface type, moisture availability, and atmospheric dynamics and thermodynamics on evaporation processes. The spatial heterogeneity of these surface types within the GSL boundary further underscores the importance of localized measurements for accurate quantification. Preliminary results indicate that the LI-710 systematically underestimates evaporation rates compared to the EC150. Feedback from LI-COR staff suggests that this underestimation is recognized and requires an LI-710 sensor firmware upgrade that will be made available during summer 2025.

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CHAPTER 1

INTRODUCTION

1.1 Motivation

Water scarcity is one of the most pressing societal issues globally. Nearly half of the world's population suffers from water insecurity for at least part of the year (Caretta et al., 2022). Arid regions are especially affected: droughts in these areas are expected to worsen, causing a further reduction in water availability (Friedrich et al., 2018). The endorheic Great Salt Lake Basin surrounding the Great Salt Lake (GSL) in Utah is one such arid region.

Terminal saline lakes such as the GSL are desiccating worldwide (Messager et al., 2016; Wurtsbaugh et al., 2017). The GSL is the largest saline lake in North America and eighth largest in the world (Wurtsbaugh et al., 2017). The volume of water in the lake has been declining since the 1980s, driven in part by natural variability and climate change but largely due to unsustainable consumptive water use (Wurtsbaugh et al., 2017; Perry, 2024). Reduced water level and newly exposed lakebed pose threats to wildlife, public health, and agriculture, among other sectors (Baxter, 2023; Bingham et al., 2023; Perry, 2024). These issues affect the nearly three million people that live along the Wasatch Front comprising roughly 82% of Utah's population (Wurtsbaugh et al., 2017).

factors, including the amount of precipitation received within the watershed, commercial and private water usage, surface and subsurface water storage in the basin, evapotranspiration (ET) from vegetation in the basin, and direct evaporation from the lake's surface and exposed lakebeds (Tanny et al., 2007; Messager et al., 2016; Wurtsbaugh et al., 2017; Friedrich et al., 2018). Evaporation is a crucial, yet understudied, component of the GSL's water balance. In the past 25 years, a roughly 8% increase in saline lake summer evaporation rates has been estimated due to increasing temperatures (Wurtsbaugh et al., 2017; Lobos-Roco et al., 2022).

At its present elevation of 1278 m, the GSL covers 45% (2514 km^2) of the 5544 km^2 below the Great Salt Lake boundary, also known as the meander line, which is located at an elevation of 1283 m above sea level. It is essential to quantify the evaporative loss from not only the lake surface, but also from mineral ponds, exposed soils, and wetlands surrounding the lake.

Mineral ponds are shallow, man-made basins designed to facilitate evaporation. Companies divert water and create areas with a high brine concentration through evaporation, which leaves behind concentrated minerals (Gwynn, 1980). These ponds are a crucial component of the mineral extraction process, allowing for the collection and refinement of minerals that precipitate out as the water evaporates (Metwally et al., 2021). There are over 340 km^2 of diked mineral ponds near the GSL, which account for roughly 6% of the total GSL area (Utah Division of Water Resources, 2024).

Currently, the exposed playa surfaces surrounding the GSL comprise roughly 1275 km^2, or 23% of the area below the GSL's meander line. The exposure of these surfaces has serious implications for public health. As more of the lakebed becomes

exposed, strong ambient winds can loft the fine playa sediments, leading to dust storms containing small-particle, heavy-metal pollutants. Such dust storms may become more frequent as the lake recedes (Perry et al., 2019; Clark and Baxter, 2023; Perry, 2024).

Wetlands constitute 26%, or roughly 1441 km^2, of the area within the GSL boundary (USGS GSL Hydro Mapper, 2024). The wetlands surrounding the GSL support large populations of migratory birds, allowing them to feed and nest in these areas (Perry et al., 2019; Baxter, 2023; Clark and Baxter, 2023). Wetlands also serve many other functions, including flood control, water quality improvement, and biogeochemical processing. Phragmites, also known as the common reed, is an invasive species that thrives in the wetlands of the GSL basin. Phragmites provide habitat and nesting areas for certain bird species, but can often outcompete native salt marsh vegetation, altering the habitat and hydrology of the wetlands.

Estimating evaporative loss within the GSL boundary as accurately as possible is important in order to help inform policies to allocate available water resources arising from precipitation, runoff and groundwater (Wurtsbaugh et al., 2017; Friedrich et al., 2018; Perry et al., 2019). Historically, evaporation data records have been collected through pan evaporation or bulk transport approximations such as the Penman-Monteith model (Shuttleworth, 2007; McMahon et al., 2013), which estimates evaporation from surface state (moisture content, land cover type and roughness) and atmospheric conditions of incoming solar radiation, temperature, vapor pressure, and wind (Tanny et al., 2007; Friedrich et al. 2018).

The difference in water vapor pressure between the surface of a water body and air above is a fundamental constraint on evaporation. For saline lakes, the saturation

vapor pressure of the water is reduced as the salinity increases (Dickson et al., 1965; Calder and Neal, 1984). For example, the saturation vapor pressure is reduced by at least 5% compared to fresh water for the present salinity of ~11% in the GSL's south arm when the water temperature is greater than 20°C, which is typically the case between May and September (Crosman and Horel, 2009). Pan measurements during summer in northern Utah suggested that typical evaporation rates for fresh water of roughly 9 mm day^{-1} decreased by as much as 60% for pan solutions approaching salt saturation, roughly 27.5% salinity (Crosman and Horel, 2009). While the lake's volume has decreased and salinity has increased from the 1980s to the present, the total mass of salt in the lake has declined by approximately 30% since its peak during the mid-1980s, due to salt deposition as the lake volume decreased (Merck and Tarboton, 2023). This loss of salt mass has implications not only for the lake's chemistry and ecological balance but also for the availability of economically valuable minerals, potentially affecting local industries that rely on mineral extraction.

Eddy covariance methods to estimate momentum, heat, and water vapor fluxes in the surface boundary layer were proposed in the 1950's but did not become practical for continuous measurements until the development of infrared gas analyzers in the 1990's (Shuttleworth, 2007; Foken et al., 2012a). Carbon dioxide and water vapor flux measurements are now made continuously at over 500 measuring sites across North and South America as part of the Ameriflux network (Novick et al., 2018) and hundreds more around the world as part of FLUXNET (Baldacchi et al., 2001).

This study is part of a pilot project funded by the Utah Division of Water Resources to support ongoing planning by the state of Utah to manage water availability

in the GSL Basin (Bingham et al., 2023). The pilot project is underway in parallel with other preliminary work by the United States Geological Survey (USGS) and Utah Geological Survey (UGS) to help plan for approaches to monitor evaporation and ET over a multi-year period for the dominant land cover types within the GSL boundary: north and south arms of the GSL, wetlands, playa, and mineral ponds. Such monitoring will be used to help validate remote sensing and modeling estimates of evaporation and ET across the entire GSL Basin. A UGS team deployed sensors during summer 2024 at a wetland site in Davis County that has been providing estimates of evaporative flux since that time. Surface and subsurface lake temperatures and basic meteorological parameters will be available from the USGS through deployment during May 2025 of two buoys in the south arm of the GSL.

1.2 Objectives

As will be discussed in this thesis, the University of Utah team was funded to deploy eddy covariance sensors over a one-year period beginning in late summer 2024 to evaluate sensor performance at a site on the Farmington Bay playa and at a second site adjacent to mineral ponds in Weber County. Similar to the UGS wetland site, each University of Utah site has two infrared gas analyzer eddy covariance sensor systems: (1) Campbell Scientific EC150 CO2/H2O Open-Path Gas Analyzer (hereafter referred to simply as EC150), which measures carbon dioxide and water vapor densities, air temperature, pressure, and three-dimensional wind speed and direction and (2) LI-COR LI-710, which measures water vapor concentration, temperature, pressure, and vertical wind only. An objective for the pilot study is to assess how well the eddy covariance

equipment can withstand the harsh playa and evaporation pond environments. The relative amount of maintenance and ancillary equipment (power, communication, etc.) required to keep those sensors functional during an entire year is also critical to evaluate.

Since the cost of a LI-710 sensor is ~20% of that of the EC150 sensor system, a fundamental objective of the pilot project is to evaluate the accuracy of the evaporation estimates from the LI-710 relative to the EC150. The performance characteristics of the EC150 as well as those of comparable 3-D eddy covariance systems such as the LI-COR LI-7500 system are well established (Munger et al., 2012; Rebmann et al., 2012). While 3-D systems have a minor drawback of underestimating fluxes when the wind direction is parallel to the sensor orientation, the flux estimates from the EC150 are viewed here as likely to be closer to truth than those from the LI-710. In addition, this pilot project is focused on assessing sensor performance for estimating evaporation, rather than the common practice to deploy 3-D eddy covariance systems along with additional sensors to estimate all terms in the surface energy budget for a locale (Foken et al., 2012b; Munger et al., 2012; Vesala et al., 2012; Massey et al., 2017).

The pilot study depends on the maintenance, communication, data collection, storage, and display infrastructure that supports the UUNET network of weather stations external to and within the GSL boundary (see https://horel.chpc.utah.edu/uunet.html). Critical steps completed and continuing to date for the pilot study are as follows:

1. Install a three-meter tower with an LI-710 sensor and EC150 system between two mineral ponds within the Compass Minerals facility in Weber County with solar power, data storage and communication equipment. As part of the UUNET network, this site is referred to as the UUCMF site and located ~3 km from an

existing 10 m tower with meteorological sensors referred to as GSLM.

2. Install a three-meter tower with an LI-710 sensor and EC150 system with solar power, data storage and communication equipment and located adjacent to a previously deployed 10 m tower on the Farmington Bay playa in Davis County. This site is referred to as the UUPYF site while the adjacent 10 m tower with extensive meteorological sensors is referred to as UUPYA.
3. Manage the UUPYF and UUCMF sites including collecting, archiving, and providing access to data in real-time from these locations throughout the pilot project period.

Within the longer timeline planned for the pilot study, the work described in this thesis addresses specific questions relying on data collected at UUCMF and UUPYF from August 2024 to March 2025. The data analysis focuses on the period from 15 October 2024 to 31 March 2025 (referred to as the cool season) when both eddy covariance systems were fully operational. Those questions are:

1. To what extent are the automated quality control flags provided by both sensor systems identifying periods of poor sensor performance as well as periods when the sensors are not providing reasonable estimates of evaporative flux?
2. Under what surface boundary layer conditions may the LI-710 and EC150 systems be providing accurate evaporation estimates over a playa surface and near mineral ponds? To what extent do those estimates differ as a function of season, time of day, and boundary layer state?
3. Under what surface boundary layer conditions are differences in evaporative flux measured by the two systems reflecting limitations of either sensor system? Are

those differences indicative of consistent biases as a function of boundary layer state that might be correctable?

The data and methods used in this study are described in the following chapter. The results in Chapter 3 address the questions described above. A summary of the work completed following in Chapter 4.

CHAPTER 2

DATA AND METHODS

This study focuses on evaporation measured at the playa and mineral pond sites. The results of this study may help validate future estimates of evaporation obtained from modeling and remote sensing approaches for such land cover types within the GSL meander line. At each of the two sites, the performance of two different infrared gas analyzer sensor systems will be compared. The LI-COR LI-710 measures vertical (1-D) wind, water vapor concentration, air temperature, and pressure, while the Campbell Scientific EC150 measures CO_2 and water vapor densities, air temperature, pressure, and three-dimensional (3-D) wind. The questions listed in the previous chapter are assessed on the basis of the comparable eddy-covariance-based sensible (*H*) and latent (LE) heat flux estimates at 30-minute intervals available from both sensor systems. Direct comparisons of the evaporation rate from the two sensor systems will be made as a function of siting and environmental conditions that include cloud cover, wind, temperature, and relative humidity.

2.1 Study Area

Data are being collected from four sites located within playa and evaporation pond surface types surrounding the GSL: GSLM/UUCMF (mineral ponds) and

UUPYA/UUPYF (playa). These sites are not entirely representative of playa and pond locales as a whole within the GSL boundary, but the locations were determined based on many pragmatic factors (access constraints, existing nearby infrastructure, homogeneity of terrain, e.g.). Throughout the duration of the pilot project, direct evaporation from the exposed soils and shallow mineral ponds will be compared at the flux sites (UUCMF and UUPYF) with supplementary meteorological data used from the nearby 10 m towers (GSLM and UUPYA). The focus in this study is on the conditions available from the installation of the flux sites through March 2025. Fig. 1 shows the area of study and location of the four sites used as part of this study. The location of the UGS wetland site (UGSPG) is also shown.

2.1.1 Mineral Ponds (UUCMF/GSLM)

Two towers are located within the Compass Minerals facility (Fig. 2): the existing 10 m tower at GSLM that has been maintained as part of the UUNET network since 2013, and the UUCMF site installed between two mineral ponds that are mined during the year. The 10 m tower at GSLM provides wind, temperature, relative humidity, pressure, incoming solar radiation, and precipitation from an unheated tipping bucket and a Lufft WS-100 micro-radar sensor.

On 15 Aug 2024, a new 3 m tower with eddy covariance sensors described previously was deployed on a berm (1.5 m tall and 10 m wide) between two mineral ponds roughly 3 km west of GSLM. The GSLM site is not appropriate for assessing evaporation because of nearby buildings and pump facilities and due to its location adjacent to an evaporation pond that is not usually mined. The UUCMF location was

chosen in coordination with Compass Minerals staff to provide data that are more representative of the conditions across the facility and specifically of the ponds immediately to its north and south. Salt-saturated water from the GSL's north arm is pumped to the northern edge of the facility and then circulates by gravity and pump stations through the mineral ponds. The pond to the north of UUCMF was allowed to dry out during Fall 2024 while the pond to the south remained covered by typically less than 10 cm of water throughout the cool season. As will be shown later, the northwest-southeast oriented berm on which the UUCMF site is located is roughly perpendicular to the common thermally-driven flows from the northeast at night and from the southwest during the day.

2.1.2 Playa (UUPYA/UUPYF)

The UUPYA 10 m tower site was installed during June 2022 by Profs. Kevin Perry and Sebastian Hoch in the Department of Atmospheric Sciences to monitor dust lofted and transported away from playa surfaces. An extensive suite of meteorological sensors, soil sensors and aerosol samplers are available from the site, including air temperature and relative humidity at 3 vertical levels, wind speed at 5 levels and wind direction at 10 m, incoming and outgoing solar and infrared radiation, 1, 2.5, 4, and 10 μm particulate concentrations, ground heat flux at 5 cm, and soil temperature, moisture content, and conductivity at multiple levels between 2 and 50 cm. The standalone UUPYF eddy covariance site was installed 20 m to the northeast of UUPYA on 26 Sep 2024. The power system located between UUPYA and UUPYF was updated at that time as well. Fig. 3 shows the UUPYA and UUPYF sites.

2.1.3 Data Storage

Table 1 provides details on four UUNET stations near the Great Salt Lake, including location, network name, start date, and data access links, with continuous publicly available data. JSON (JavaScript Object Notation) is relied upon to store, read, and analyze data from the flux sensors, as there is output available from over 200 variables from each pair of sensors. JSON's structured format allows these data to be stored and visualized uniformly, with detailed sensor diagnostics displayed alongside flux data. The JSON files have facilitated evaluation and analysis of sensor performance and periods of unrealistic flux estimates, as well as seasonal, diurnal, and boundary-layer-dependent differences between the two sensor systems.

2.2 Sensor and Station Installation

2.2.1 Wind Analysis

Prior to deploying towers at the two stations, predominant flows throughout the year were assessed at each site in order to orient the eddy covariance sensors. To illustrate differences in flow between the sites, Fig. 4 displays the variability in wind speed and direction during the month of July 2024, prior to deployment. GSLM (Fig. 4a) displays a dominant nighttime northeasterly flow and daytime southwesterly flow during summer when evaporation is greatest. The EC-150 and LI-710 sensors were mounted on a tower support arm and directed to the northwest to be positioned perpendicular to this flow. UUPYA has a dominant WNW flow during the day and ESE flow at night (Fig. 4b). Accordingly, infrared gas analyzer sensor systems were mounted on a tower arm pointing northeast to capture the predominant northwest-southeast flows.

2.2.2 Flux Footprints

Many factors must be considered when siting eddy covariance systems, including instrument height, sensor separation, surface heterogeneity, and the influence of nearby obstacles or non-representative land covers (Munger et al., 2012). One of the most important considerations is the flux footprint – the upwind area that contributes to the measured fluxes at the sensor (Rannik et al., 2012). The flux footprint is not static; it varies substantially with micrometeorological conditions, particularly atmospheric stability and the intensity of turbulence within the boundary layer (Burba, 2005; Rannik et al., 2012).

The size and shape of the footprint depend strongly on atmospheric stability. Under unstable conditions, enhanced turbulent mixing typically shortens the footprint such that fluxes are influenced by a smaller upwind area. If the upwind land surface type is heterogeneous, a reduced footprint is beneficial to integrate the fluxes across similar land cover types close to the sensor. In contrast, under stable conditions, turbulence is suppressed, and the footprint becomes more elongated farther upwind along prevailing wind directions. Thus, the footprint generally increases as the stability transitions from unstable to neutral to stable, reflecting weaker vertical mixing and longer transport distances (Kljun et al., 2004; Rannik et al., 2012).

Prior to installation at UUCMF and UUPYF, subjective assessments of the likely flux footprints under typical wind and stability conditions were made to ensure that the sensors captured fluxes representative of their target surface types – mineral ponds at UUCMF and exposed playa at UUPYF. Given the uniformity of the underlying land surface types at these sites, variations in the footprint from which the fluxes were

originating were expected to be less important. However, UUCMF was sited preferentially on the southern edge of the berm to minimize what was expected to be greater evaporation from the southern pond during summer afternoons. Estimates of the footprints at the two sites during the cool season will be presented in Section 3.3.4.

2.2.3 Deployment

Site installation for UUCMF was completed on August 15, 2024, and UUPYF was deployed on September 26, 2024. The deployments began early in the morning to avoid high afternoon temperatures. While UUCMF is accessible by vehicle, UUPYF requires a 3 km bike ride or walk in each direction to access the site from the nearest road. In addition, UUPYF is often inaccessible during winter and spring, as the area may be flooded or too muddy. Access is only practical when the surface is dry or frozen.

Equipment required at each site other than the eddy covariance sensors includes the tower on which to mount sensors, stakes for securing the tower to the ground, solar panels, and wooden platform for mounting solar panels and the waterproof enclosure for the CR1000X Campbell Scientific datalogger, cell modem, and batteries.

2.3 Eddy Covariance Method

Eddy covariance measures and calculates vertical turbulent fluxes within the Earth's atmosphere to make estimates of greenhouse gas (GHG) emissions and evaporation (Rebmann et al., 2012). It does so through correlating vertical wind velocity with gas concentration (primarily H_2O and CO_2). Eddy covariance flux calculations are made through a Reynolds decomposition of vertical flux. Vertical flux is shown as

follows:

$$F = \overline{\rho_d w s}$$

The flux is the averaged product of the following variables:

ρ_d = dry air density

w = vertical wind speed

s = dry mole fraction of the gas of interest

The Reynolds decomposition separates mean and instantaneous deviations from the mean for each variable:

$$F = \overline{(\overline{\rho_d} + \rho_d')(\overline{w} + w')(\overline{s} + s')}$$

The full decomposition leads to multiple quadratic terms, but the equation is commonly simplified based on a few key assumptions:

1. Fluctuations in air density (ρ_d') are negligible, which is valid over flat terrain and under typical atmospheric conditions.
2. Mean vertical wind speed ($\overline{w}$) is negligible, which assumes horizontal homogeneous terrain with no large-scale vertical motion.
3. Mean scalar fluctuations ($\overline{s'}$) are zero, since fluctuations are defined as deviations from the mean and average to zero over the sampling interval.

While the first two reflect assumptions about the physical environment, the appropriateness of the third depends on the sampling interval used for the analysis. Removing these terms results in the eddy flux to be simplified as:

$$F \approx \overline{\rho_d}\ \overline{w's'}$$

Accurate eddy covariance calculations rely on certain atmospheric conditions to be present: turbulence must be fully developed and consistent over the averaging period, and the atmospheric surface layer remains statistically stationary. These constraints are typically evaluated using measures of relative non-stationarity and boundary layer stability that will be discussed in Chapter 3.

2.4 Eddy Covariance Sensors

Two eddy covariance sensor technologies are compared at each site to assess differences in their performance. Prior to the deployment of UUCMF and UUPYF, predominant flows were determined from wind data from GSLM and UUPYA to identify optimal orientations for the eddy covariance sensors. Based on the GSLM winds and the berm orientation, the eddy covariance sensors were mounted pointing 19^{o} north of west at UUCMF and 30^{o} east of north at UUPYF. The power system and 10 m UUPYA tower located 10 m and 20 m away, respectively, face the opposite direction of the sensors.

The open path EC150 sensor system from Campbell Scientific is composed of multiple components, including the EC150 IRGA and CSAT3A sensors (Campbell Scientific, 2017). The EC150 (cylinders in Fig. 5a) measures CO_2 density and H_2O density while the CSAT3A (three-prong sensors at top and bottom in Fig. 5a) measure wind speed in three cartesian directions and sonic temperature. Additional sensors (not shown) are used to measure ambient temperature and pressure.

The EC150 measures gas concentration by generating infrared radiation in the upper arm and sending it down a 15cm optical path to a mercury cadmium telluride detector in the lower arm. As the radiation travels along the optical path, carbon dioxide

and water vapor will absorb radiation at different wavelengths. The detector measures the decrease in radiation intensity due to absorption and uses the Beer-Lambert Law to calculate concentration at 10 Hz:

$$P = P_o e^{-\varepsilon c l}$$

Where:

P = irradiance after passing through the optical path

P_o = initial irradiance, ε is molar absorptivity, c is analyte concentration, and

l = path length.

A limitation of this sensor system is that the sensors must be oriented perpendicular to the dominant axis of turbulent flows, with less representative winds and gas fluxes when the flow is towards or away from the sensor housing and tower (Campbell Scientific, 2024). The impact of this limitation is evaluated in Chapter 3.

EasyFlux DL is an open-source, 10,000+ line CRBasic program developed by Campbell Scientific and uploaded to the CR1000X datalogger to calculate momentum, H, and LE, as well as evaporation rates, from the raw 10 Hz output of the EC150 sensor system. The code implements a suite of well-established corrections recommended by turbulence researchers (Aubinet et al., 2012; Foken et al., 2012b; Rebmann et al., 2012). These include coordinate rotation, time lag compensation, frequency response correction, air density (WPL) correction, spike detection, and stationarity checks. This study relies on EasyFlux to process raw high-frequency data in real time, applying the corrections summarized in the aforementioned studies (particularly Foken et al., 2012b) to produce 30-minute flux estimates (Campbell Scientific, 2024). The original 10 Hz data are

retrieved from the datalogger during site visits and archived for post-processing not included in this study.

The 1-D LI-COR LI-710 ET sensor system determines vertical fluxes of *H* and LE over 30-minute samples. The LI-710 has a pair of vertically aligned transducers that measure vertical wind speed, temperature, and pressure, along with an air intake tube that draws sampled air into a pump where water vapor concentration is measured by a fast-response relative humidity sensor. Its lower cost in comparison to 3-D infrared gas analyzer systems (e.g., EC150 and LI-COR's LI-7500) is particularly relevant given the Division of Natural Resource's need to validate evaporation estimates from models and remote sensing in order to reduce uncertainties in evaporation across the entire GSL Basin (Bingham et al., 2023). However, concerns exist about the LI-710 sensor robustness and sampling methodology. For example, since the LI-710 does not measure horizontal wind speed and direction, it is not possible to implement many of the flux corrections recommended by Foken et al. (2004, 2012b) and implemented for the LI-7500 and EC-150 (Burba, 2021; Campbell Scientific, 2024). These limitations are likely less critical for applications near the GSL since the LI-710 sensors were deployed at sites where the flux footprint is relatively uniform (LI-COR, 2024).

Table 1: UUNET station information.
Data collected continuously from all four stations are publicly available as is done for all UUNET sites. Table 1 shows information about each station.

Location	**Network**	**Name**	**Period of Record**	**UUNET link**
GSLM	UUNET	Compass Minerals	April 2013 –	https://mesowest.utah.edu/cgi-bin/droman/meso_base_dyn.cgi?stn=gslm
UUCMF	UUNET	Compass Minerals Flux	Aug 2024 –	https://mesowest.utah.edu/cgi-bin/droman/meso_base_dyn.cgi?stn=UUCMF
UUPYA	UUNET	Playa	June 2022 –	https://mesowest.utah.edu/cgi-bin/droman/meso_base_dyn.cgi?stn=UUPYA
UUPYF	UUNET	Playa Flux	Sep 2024 –	https://mesowest.utah.edu/cgi-bin/droman/meso_base_dyn.cgi?stn=UUPYF

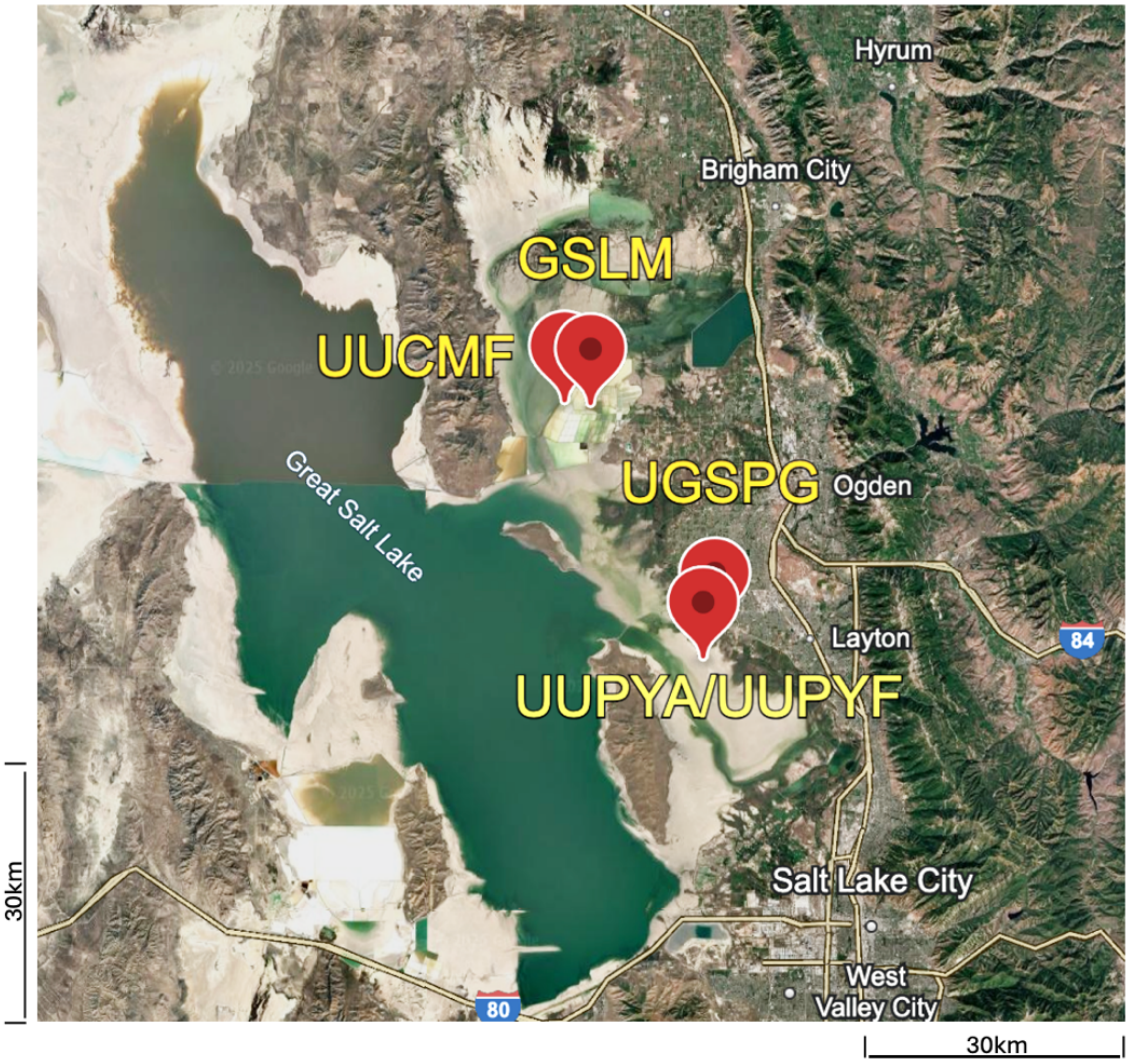


Figure 1. Locations of the pilot study flux sites (UUCMF and UUPYF), UUNET sites (GSLM and UUPYA) and UGS phragmite site (UGSPG).

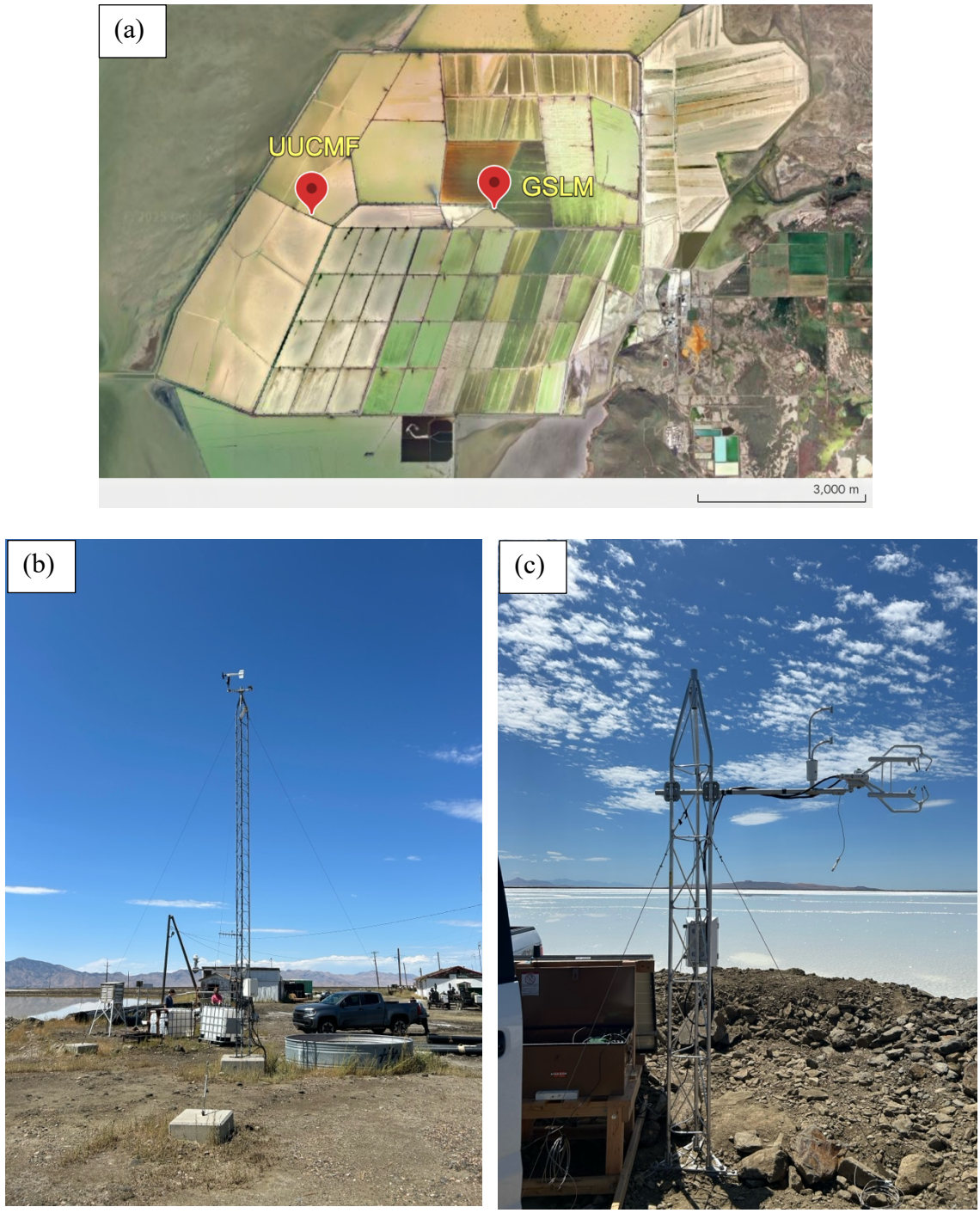


Figure 2. UUCMF and GSLM site configuration. (a) Location of the UUCMF and GSLM sites; (b) GSLM looking northwest and (c) UUCMF looking south.

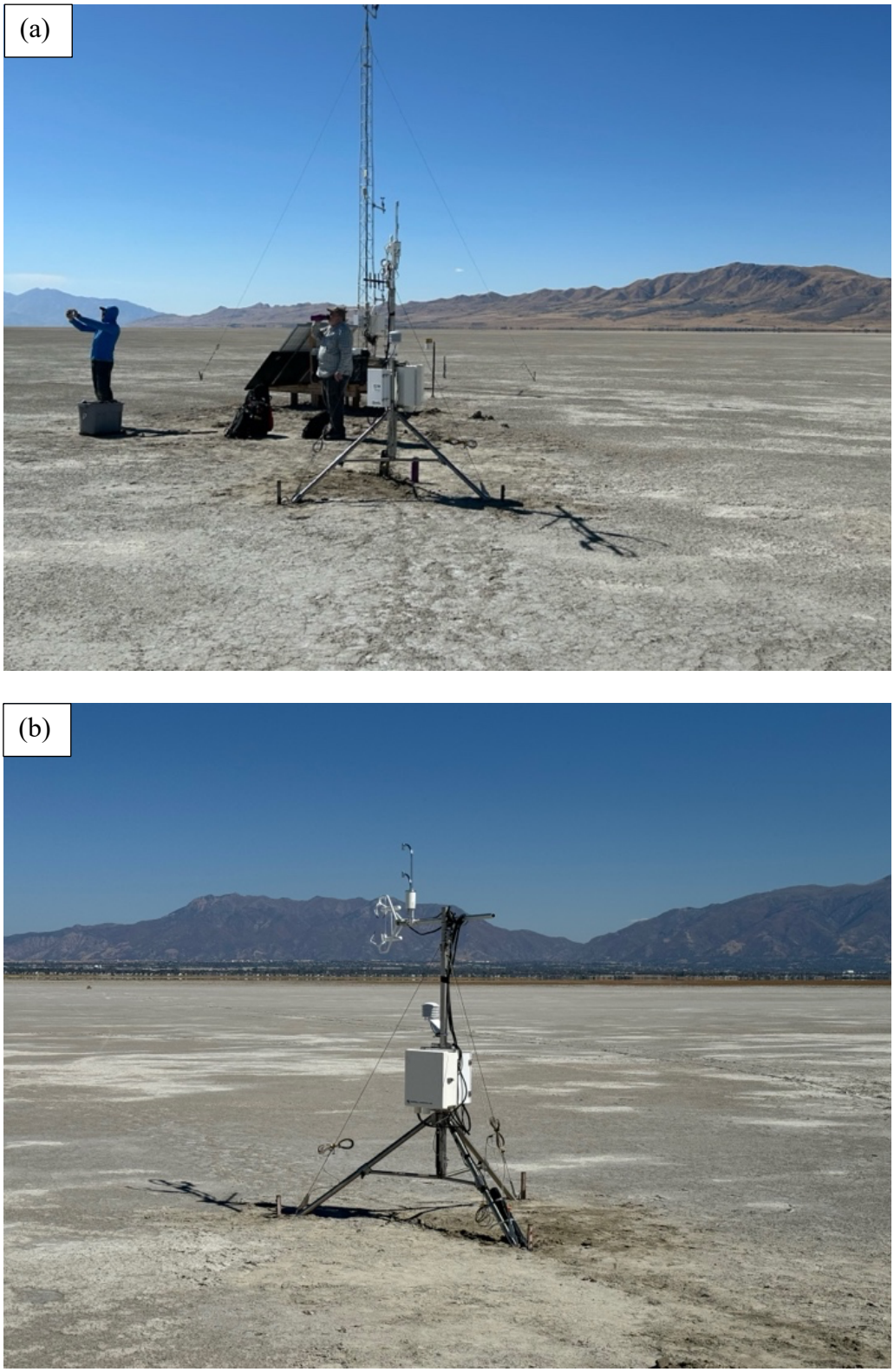


Figure 3. UUPYA and UUPYF site configuration. (a) UUPYA site and mount for the solar panels (background) and UUPYF (foreground) towers facing southwest and (b) UUPYF tower with eddy covariance sensors facing northeast.

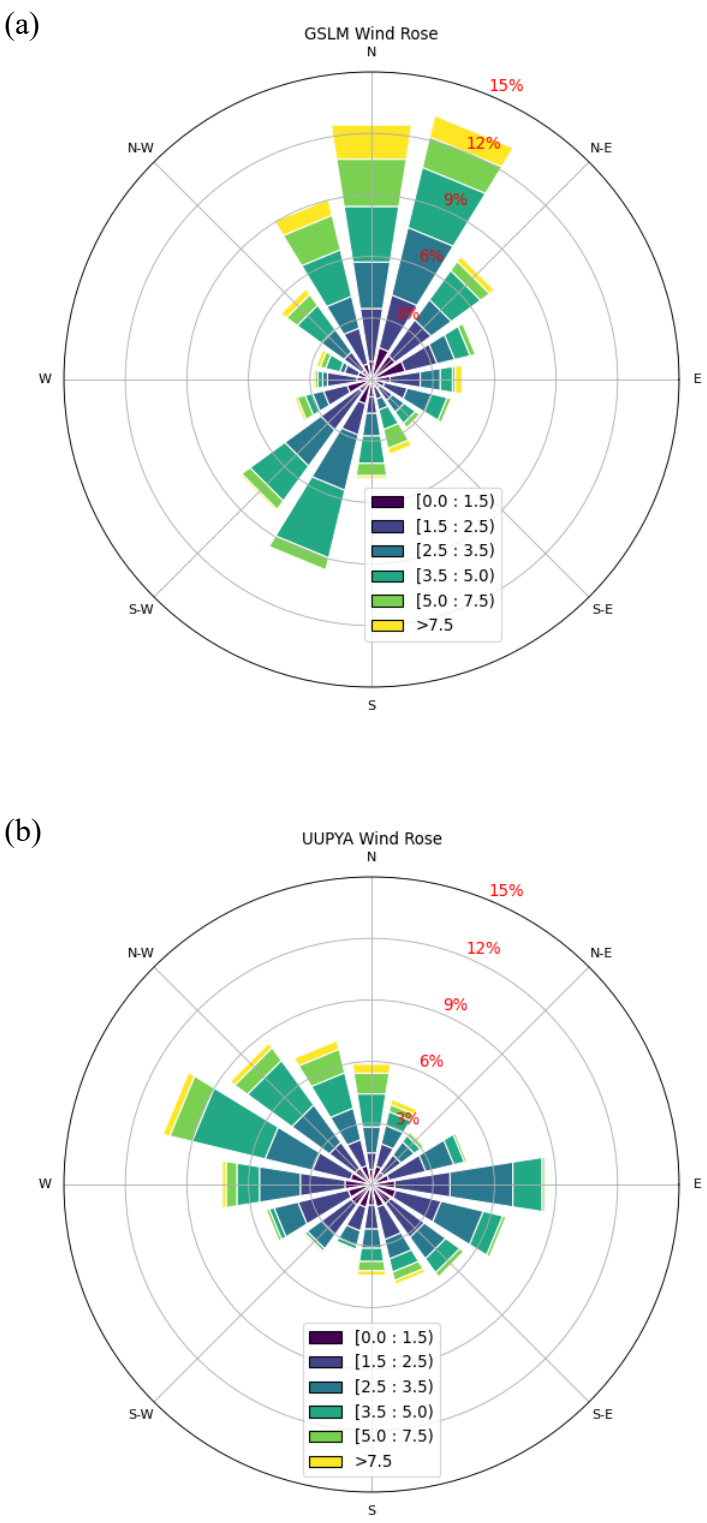


Figure 4. July 2024 wind roses (m s^{-1}) showing proportion and intensity of winds from 16 cardinal directions at both sites. (a) GSLM wind rose and (b) UUPYA wind rose.

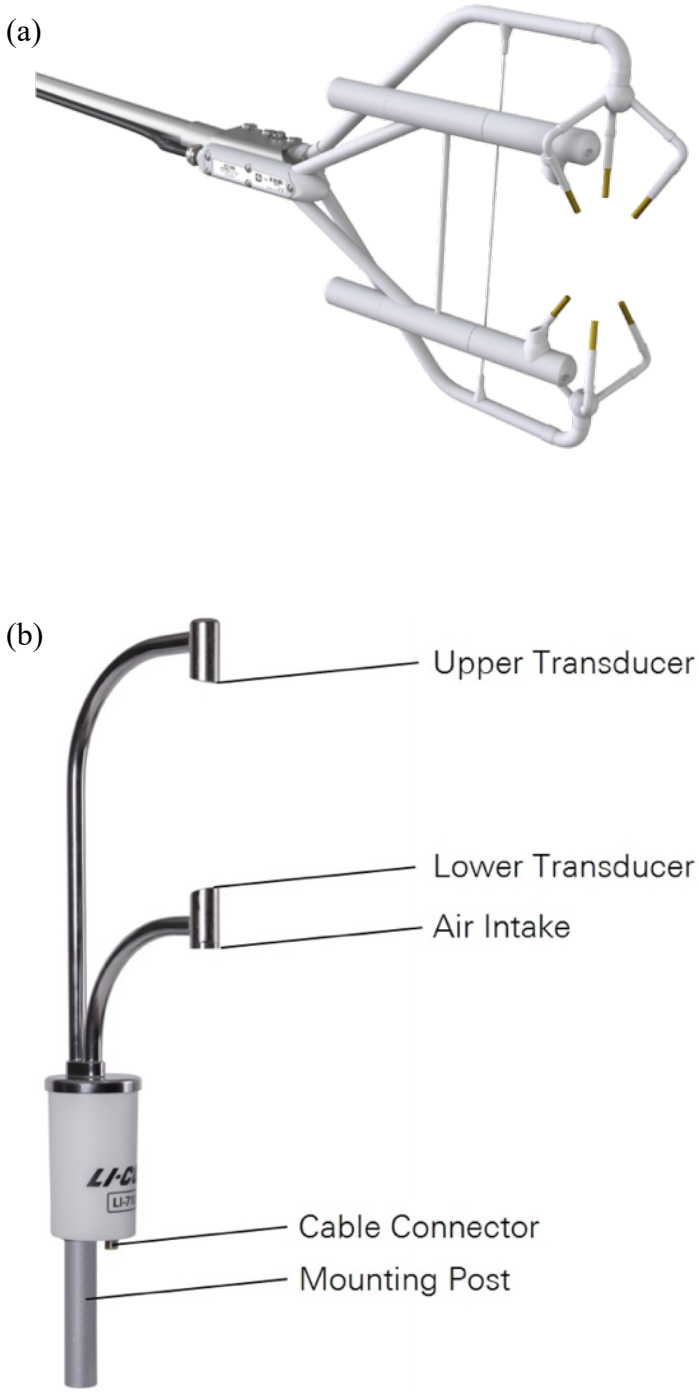


Figure 5. Eddy covariance sensors used during the project. (a) The Campbell Scientific EC150/CSAT3 sensor system, priced at $30,000 and (b) The LI-COR LI-710 Sensor, priced at $6000.

CHAPTER 3

RESULTS

3.1 Site logs

As shown in Fig. 6, seasonal temperature and wind speed variations, eddy covariance sensor performance issues, and site outages have been tracked as part of quality control (QC) procedures. Relative to the prior winter, temperatures tended to be mild with high winds accompanying only occasional storms. Sensor issues arising from environmental factors such as ice fog are more frequent than equipment failure or system communication outages. Site and sensor outage tracking has been used to help identify and remove unreasonable data from the dataset used for analysis.

3.2 Sentinel Hub Satellite Imagery

Satellite imagery available from the Copernicus Data Space Ecosystem Browser provides broad coverage during cloud-free periods to analyze environmental factors influencing evaporation at the playa and mineral pond sites. Sentinel-2 imagery provides 10-meter resolution, allowing a detailed analysis of water boundaries, and the wide swath coverage ensures that the entire GSL can be monitored consistently. With 2–3 overpasses every five days for a given location, Sentinel-2 provides frequent observations that support tracking short-term environmental variability. Sentinel-2 imagery facilitates

monitoring environmental conditions such as cloud cover and type (water versus ice clouds) through short wave infrared (SWIR) images. The extent of surface water is visualized through the normalized difference water index (NDWI), and land cover moisture is estimated from the normalized difference moisture index (NDMI). The NDWI imagery as shown in Fig. 7 allows tracking fluctuations in the lake's surface area, indicating water loss due to evaporation over time (Craft and Horel 2019).

Sentinel-2 satellite imagery helps visualize water coverage at each station during the pilot study period. This facilitates an analysis of evaporation based on varying ground and vegetation moisture conditions. The variations in surface water at UUCMF and UUPYF in the camera images in Fig. 7 help to interpret the Sentinel-2 images on those days. Knowing the dates of these images makes it possible to revisit the evaporation data and examine how evaporation may have varied under different environmental conditions.

3.3 Sensor Data QC

The analysis in this study focuses on the period from mid-October 2024 through March 2025. While sensors were installed earlier at UUCMF and UUPYF, the LI-710's fast-response relative humidity sensors at both sites during that period occasionally reported near-zero humidity readings and hence, inflated LE and evaporation rate estimates. After consultation with LI-COR staff, replacement sensors were provided and installed on 15 October at UUPYF and 16 October at UUCMF. All subsequent data were subject to rigorous quality control, and only observations from 16 October 2024 through 31 March 2025 are used in the flux comparison and evaporation analyses presented in this study. This period will be referred to as the "cool season" as measurement will

continue through October 2025 in order to capture the sensor performance through the "warm season" and subsequently cover an entire annual cycle.

In addition to the faulty relative humidity sensor, a number of issues have come to light throughout the duration of this project. First, the sensors are sensitive to the elements, and this has led to periods of compromised data. Some examples are spiderwebs and bugs blocking the LI-710 intake at UUCMF during the summer, rodents chewing through wires, rain and snowstorms, and stable conditions with freezing fog during winter. Fortunately, the 2024-2025 winter season was relatively mild (Fig. 6) as conditions at the UUPYA site during other winters were often colder, with water, snow and ice having even greater impacts. The fog conditions on 5 December evident in Fig. 8 illustrate one type of environmental condition that affects ET sensor performance. Additionally, the internal cell temperature of the LI-710 is recommended to be kept above 5°C to avoid damaging the pump. Insulating the LI-710s at UUCMF and UUPYF with pipe wrap (Fig. 9) kept the cell above this threshold approximately 97% of the time at both sites, even when the ambient temperature was as low as -10°C. All of these factors were considered when evaluating the data.

Quality Control (QC) checks are necessary to determine which 30-minute measurement intervals had reliable data from both sensors. This step involved identifying when vendor-supplied real-time diagnostic codes report neither, one, or both sensors as having potentially erroneous data. In addition to sensor performance indicators, the Campbell Scientific EasyFlux software for the EC150 provides a QC flagging system based on wind and turbulence characteristics to assign a numerical grade from '1' (good) to '9' (likely erroneous) to each flux measurement. These QC values are derived from

multiples tests, which include tests for stationarity and integral turbulence characteristics as described by Foken et al. (2004, 2012b), as well as from wind direction measurements. QC values are available from the EC150 for the fluxes of interest to this study:

- $\overline{w'u'}$: momentum flux (τ)
- $\overline{w'T'}$: sensible heat flux (H)
- $\overline{w'q'}$: latent heat flux (LE)

QC values for evaporation rate are identical to those for LE since the evaporation rate is simply the LE divided by the latent heat of vaporization.

Three of the many tests completed in real time for each 30-minute sample by the EasyFlux software are now summarized.

3.3.1 Wind Direction

In accordance with the guidelines from Campbell Scientific, wind coming from angles of 180° ± 10° relative to the sonic coordinate system results in a QC value of '3' and should be ignored, as the wind flowing parallel to the sensors is altered by the sensor configuration. Wind direction within 29° of 180° results in QC values of '2' and should be treated with caution. Because of the initial wind analysis mentioned in Chapter 2, the sensors were mounted in directions that minimized winds from these directions, as summarized in Table 2. Hence, this potential limitation is not a substantive problem for this study.

Fig. 10 shows the variation in wind at each site over the entire collection period. Most notably, the southerly flows at UUCMF present during the summer are reduced during the cool season. These seasonal differences are driven by the dominant mountain–

valley circulations in the region. During winter, nighttime differential cooling over land and water enhances down valley flows, leading to persistent northeasterly winds draining toward the GSL through the Bear River valley. Daytime heating is also weaker than in summer, resulting in less pronounced flows up-valley and away from the lake. These circulations creates a seasonal asymmetry in the diurnal wind cycle, with stronger and more persistent flow toward the lake during the day during winter than is common during the summer. The direction of the winter flows at UUPYF are more consistent with those observed during summer, but, as might be expected, the fraction of both very light and high wind speeds are increased during winter due to more frequent stable periods and storms, respectively.

3.3.2 Turbulence Characteristics

Identifying turbulence characteristics is a critical part of the EC150's QC system, in order to assess whether conditions during each 30-minute averaging interval are suitable for reliable eddy covariance measurements. The two main factors contributing to the turbulence-based QC output are relative non-stationarity and boundary layer stability.

3.3.2(a) Relative Non-Stationarity

Stationarity refers to the assumption that atmospheric conditions remain statistically constant over the averaging period, here 30 minutes (Campbell Scientific, 2022). The eddy covariance method assumes that turbulent fluxes remain similar throughout the measurement interval, otherwise the 30-minute sample means of vertical wind and water vapor concentration may be affected by sub-sample turbulent variations

that in turn affect the flux estimates. A relative non-stationarity test is applied that computes the ratio of the variance among the six 5-minute sub-sample flux estimates relative to the 30-minute flux estimate (Foken et al., 2004; Foken et al., 2012).

If the variability between the 5-min fluxes within a 30-minute interval is high (ratio greater than one relative to the 30-minute sample), then weather conditions may be evolving due to frontal passages or thermally-induced wind shifts. However, if the turbulence within the 30-minute sample is very weak, which is common during stable periods, then the stationary test flags such periods with a high QC value. For this work during the cool season, flux estimates are accepted if the non-stationarity test returns a value of 5 or less, which corresponds to ratios of 1 or less (Campbell Scientific, 2024). As will be discussed later, higher values were common during cool season stable periods, particularly during early morning or nighttime periods and more frequently at the Playa site (UUPYF) than at the Compass Minerals site (UUCMF).

3.3.2(b) Boundary Layer Stability

A key parameter used to evaluate atmospheric stability is the Monin–Obukhov stability ratio, expressed as $\zeta = Z/L$, where Z is the aerodynamic measurement height and L is the Monin–Obukhov length scale. Z is calculated as the sensor height above ground minus the zero-plane displacement height. At the flat playa site, Z is set to 2.3 m, corresponding to the sensor height. At the UUCMF site, although the sensor is mounted 3.3 m above the mineral pond surface due to a 1 m tall berm, Z is also set to 2.3 m to account for the influence of the berm on surface-layer dynamics.

Changes in the stability ratio ζ in this study depend entirely on the Monin-Obukhov length scale, L. That length scale depends on the relative influence of vertical virtual temperature flux ($\overline{w'\theta_v'}$) and momentum flux, ($\overline{u'w'}$) (Stull, 1988):

$$L \propto -\frac{\left|\overline{u'w'}\right|^{3/2}}{\overline{w'\theta_v'}}$$

When the temperature of the underlying surface is higher than the air above (positively buoyant conditions), then the denominator will be positive in the presence of turbulence. As the buoyancy increases then for a given momentum flux, L decreases and the stability ratio ζ becomes increasingly more negative. When the air temperature aloft is higher than the surface temperature (stable conditions), then the denominator will be negative. As the turbulence and vertical momentum flux weakens during a stable period, then ζ becomes increasingly more positive. Summarizing these cases and assuming that the surface vapor pressure is higher than the atmospheric vapor pressure:

- $\zeta > 0.4 \rightarrow$ Very stable conditions that likely inhibit LE (e.g., at night or over cold surfaces)
- $0.4 > \zeta > 0 \rightarrow$ Stable conditions: LE likely to be weak
- $\zeta \sim 0 \rightarrow$ Neutral conditions: turbulence is driven by wind shear, H is small, and LE likely much higher than during stable periods
- $-0.4 < \zeta < 0 \rightarrow$ Unstable conditions: buoyancy enhances turbulence with increased LE (e.g., during afternoon)
- $\zeta < -0.4 \rightarrow$ Very unstable conditions: H and LE may be poorly estimated due to the irregularity of the turbulence during the 30-minute samples.

Table 3 summarizes that UUPYF tended to have more frequent stable and very unstable conditions. The percentages do not add up to exactly 100% due to a small number of anomalous values removed due to sensor errors.

The ζ rose plots in Fig. 11 highlight the more frequent stable conditions at UUPYF than at UUCMF. Later results will further emphasize that monitoring evaporative fluxes at UUPYF during the cool season is more difficult than at UUCMF. Boundary layer stability at the two sites during the cool season reveals atmospheric differences that reflect surface type and prevailing meteorological conditions. As shown in the *Z/L* wind roses in Fig. 11 and Table 3, UUPYF experiences a wider range of stability states, with a significantly greater frequency of both highly stable conditions ($Z/L \geq 0.4$, 9.2%) and highly unstable conditions ($Z/L \leq -0.4$, 16.1%) compared to UUCMF (4.4% and 10.0%, respectively). This broader distribution at UUPYF is characteristic of the exposed playa surface, which undergoes more extreme diurnal temperature fluctuations and transitions between dry, frozen, and flooded states. In contrast, UUCMF remains predominantly neutral to slightly stable, with 85.2% of values falling within the $-0.4 < Z/L < 0.4$ range. This stability pattern reflects the thermal inertia of the mineral ponds and persistent northerly flow down the Bear River drainage.

The UUPYF *Z/L* wind rose in Fig. 11b shows that highly unstable conditions (darker bins) are concentrated during the daytime hours, associated with westerly to northwesterly flow. This diurnal signal indicates periods of surface heating, which enhances buoyant turbulence and results in more frequent breakdowns of the stable boundary layer. In contrast, unstable conditions at UUCMF are less frequent and more

directionally constrained, reinforcing the interpretation that the mineral ponds experience more neutral stability conditions overall.

3.3.3 Diagnostic Code Summary

The grades recommended by Campbell Scientific (2024) for each 30-minute momentum, *H*, and LE estimate from the EC150 are determined from the three tests described in the previous subsection following Foken et al. (2012b) and summarized in Table 4. Those resources define the ranking '1' as best and '9' as worst. Based on the results obtained to date from UUCMF and UUPYF, grades ≤ 6 indicate acceptable conditions for further analysis, while values from 7-9 represent questionable data that likely need to be discarded (Foken et al., 2012b).

LI-710 diagnostics relate primarily to sensor performance with a grade of 0 indicating normal operations, while increasingly larger nonzero values suggest one or more hardware or environmental issue affecting sensor performance – such as airflow blockage, internal condensation, or the sensor cell temperature falling below safe operating thresholds (LI-COR, 2025).

To compare sensor performance effectively in this study, only 30-minute intervals where the EC150 reported no sensor issues, EC150 grade values were less than seven for all fluxes, and the LI-710 returned a diagnostic of 0 were used in the following analyses. This dual-filtering is intended to reduce discrepancies between the sensor flux estimates that may have arisen from sensor malfunction or environmental factors for which the sensors are not designed.

Figs. 12 and 13 and Table 5 summarize the distribution of EC150 Grades at each site during the cool season. UUPYF exhibits a substantially higher portion of high Grades across all variables than UUCMF. No noticeable trend within the cool season is shown in Fig. 13, with the exception of fewer accepted QC grades in December due to more frequent periods of high atmospheric stability and thus higher frequency of QC > 6.

3.3.4 Flux Footprint

Kljun et al. (2004) developed a widely used approach and code to estimate the location and relative importance of passive scalar sources influencing flux measurements. These footprint estimates depend on receptor height, atmospheric stability, and surface roughness. They used a scaling procedure over a range of stratifications from convective to stable and a parameterization for the influence of roughness length to estimate flux footprints.

After minor code revisions to speed up the most recent version of the Kljun et al. (2004) python code, Figs. 14a and 14b show the flux footprints using all complete 30-minute samples from the EC150s during the cool season that pass Kljun et al. (2004) criteria to eliminate low friction velocity ($u^* < 0.1$ m s^{-1}, i.e., limited vertical momentum flux common during stable periods) or excessive instability ($\zeta > -15$). Those thresholds removed 24% (50%) of the 8736 (7890) 30-minute complete samples at UUCMF (UUPYF). These reductions are due primarily to the friction velocity constraint.

Figs. 14c and 14d illustrate the impact of removing additional samples with EC-150 grades of seven or higher that led to removal of 5% (9%) of additional samples at UUCMF (UUPYF). The slight differences between the upper and lower panels reflect

primarily ignoring samples from wind directions within ±29° of the sensor orientation. If the large number of flux estimates during high stability periods (low friction velocity) during the cool season were included, then the footprint would likely extend further away from the sensor location (Burba, 2005; Rannik et al., 2012). Hence, the objective is met at both sites to have the flux estimates reflect the conditions over the relatively homogeneous mineral pond and playa surfaces.

3.4 Data Analysis

3.4.1 Quality Control Steps

For the following analyses, a series of quality control checks were implemented. First, data prior to 16 October 2024 was not considered, as there were firmware issues with the LI-710 sensor at both UUPYF and UUCMF resolved on that day. Next, any 30-minute samples with *H*, LE, or momentum flux grades greater than '6' from the EC150 were filtered out. Then any LI-710 samples with non-zero diagnostic values were discarded. After these checks were complete, a few 30-minute *H* and LE raw values greater than 200 W m^{-2} were not considered as such values are unreasonable during the cool season. Finally, data were merged from the EC150 and LI-710 sensors for the same 30-minute samples and only those samples with complete data are used in the sensor comparisons.

3.4.2 Clear-Sky Case Study

To evaluate sensor agreement under typical operating conditions, a case study was conducted for 24 March 2025, a day characterized by mostly clear skies and sensor

function with no issues at both sites. This date was selected to represent a "best-case" scenario for intercomparison of flux estimates under favorable meteorological and radiative conditions.

Fig. 15 displays the diurnal cycle of incoming shortwave radiation at the GSLM and UUPYA sites. At UUPYA (Fig. 16b), solar radiation followed a smooth, parabolic curve typical of clear-sky days, with peak values near solar noon and little variability throughout the daylight period. In contrast, GSLM (Fig. 16a) showed fluctuations in shortwave radiation during the afternoon, indicating intermittent high cloud cover.

Sentinel-2 imagery from 24 March indicates that the pond to the south of UUCMF was filled with brine while the northern pond was partially desiccated (Fig. 16a). In contrast, the imagery near UUPYF (Fig. 16b) indicates muddy and/or dry surfaces.

Wind direction also plays a critical role in determining the surface characteristics that contribute to measured fluxes. Since the footprint of flux observations shifts with wind, the upwind land cover – whether brine-filled pond, desiccated salt crust, or muddy playa – influences the energy balance and flux measurements. These directional effects, along with differences in temperature and wind speed, are visualized in Fig. 17. UUCMF experienced relatively steady southerly winds and moderate diurnal temperature variations, while UUPYF showed greater temperature swings and more variable wind direction, with generally weaker wind speeds indicative of reduced turbulent mixing.

At UUCMF, air temperatures from both the EC150 and LI-710 sensors ranged from ~9 °C at 08:00 to ~17 °C at 19:00. UUPYF exhibited a slightly larger diurnal temperature range, with early morning lows below 7 °C and afternoon highs nearing

17 °C. These differences suggest that UUPYF experiences stronger surface cooling overnight, likely due to lower surface moisture and the higher heat capacity of the brine-filled pond at UUCMF moderating temperature swings.

Wind speed and direction patterns further distinguish the two sites. At the mineral pond site, wind speeds remained relatively steady throughout the day, generally ranging between 2-6 m s^{-1}, with wind direction predominantly out of the south and southeast. This consistent flow likely resulted in a fairly uniform fetch from the south pond, which was filled with brine. In contrast, the playa site experienced lower overall wind speeds, typically between 1-3 m s^{-1}, with brief gusts reaching 6 m s^{-1} near midday. Wind direction at UUPYF was more variable but generally followed a diurnal pattern consistent with thermally driven circulations: SSE flow at night gradually shifting toward WNW flow during the day. This shift, along with the lower wind speeds, suggests weaker turbulent mixing at the playa site compared to the mineral ponds, particularly during the morning and evening transition periods. These differences in temperature and wind are likely to influence the strength and timing of turbulent fluxes at both sites, discussed below.

Wind roses for 24 March 2025 (Fig. 18) reinforce the wind patterns described above. At UUCMF, winds were predominantly from the south (over 30% of observations), with frequent speeds exceeding 2.5 m s^{-1}, including more than 10% above 5.0 m s^{-1}. This persistent southerly flow supported sustained turbulence, as reflected in friction velocity (Fig. 19c), which remained above 0.1 m s^{-1} throughout the day and peaked around 0.31 m s^{-1}. In contrast, UUPYF experienced more variable wind directions and weaker speeds.

QC Grades (Fig. 19a) further contextualize the data reliability. At UUCMF, 30-minute LE QC Grades generally remained below 7 except for a few early-morning and late-evening periods, suggesting high-confidence flux measurements across most of the day. In contrast, UUPYF had many half-hour periods graded 8 or 9 before 11:00 and after 18:00 MST, with only the core afternoon window (11:00–18:00) containing Grade <7 data suitable for analysis under strict QC filters.

The Monin-Obukhov stability parameter ζ (Fig. 19b) further illustrates the contrast in atmospheric mixing between the two sites on 24 March. During the day, both UUCMF and UUPYF experienced unstable conditions ($\zeta < 0$), supporting buoyancy-driven turbulence. At UUCMF, ζ remained between approximately −0.2 and −1 throughout the daylight hours, indicating persistent but moderate instability conducive to sustained turbulent fluxes. At UUPYF, ζ dropped to more negative values in the afternoon (reaching −4 to −7), suggesting periods of stronger localized instability, possibly tied to patchy surface heating over the drier, darker playa. At night, the sites diverged: UUCMF remained slightly unstable with ζ hovering between 0 and −0.2, while UUPYF transitioned into weakly stable conditions ($\zeta \approx 0$ to +0.7). These differences are consistent with nighttime surface cooling being more pronounced at the playa and further support the notion of stronger and more persistent turbulence at UUCMF.

Friction velocity at UUPYF (Fig. 19c) stayed below 0.1 m s^{-1} for much of the day except between 11:00 and 18:00 MST, indicating more limited periods of effective turbulent transport, which is consistent with the higher number of rejected footprint samples at UUPYF during the cool season. The $u^* \leq 0.1$ m s^{-1} threshold follows EasyFlux DL guidance (Kljun et al. 2004), while more conservative criteria (e.g., $u^* \leq$

0.2 m s^{-1}, per LI-COR and Campbell Scientific) would exclude many additional low-turbulence intervals.

The combined influences of wind direction, turbulence intensity, and QC confidence grades help explain the flux patterns observed in Fig. 20. At UUCMF (Fig. 20a), *H* estimates from the EC150 and LI-710 were in close agreement for much of the day, both peaking around 16:00 MST at approximately 60 W m^{-2}. However, just after 15:00 MST, the two sensors diverged: the EC150 recorded a sharper and more pronounced spike in *H* compared to the LI-710. This discrepancy corresponds to a period when wind direction shifted to flow nearly parallel to the EC150's mounting orientation, likely causing distorted airflow and elevated QC flags. In contrast, LE estimates between the sensors differed throughout the afternoon. The EC150 reported peak 30-minute average LE values of ~90 W m^{-2} around 20:00 MST, while the LI-710 peaked earlier and lower, near 70 W m^{-2} at 18:00 MST. This sustained mismatch in LE, despite general agreement in *H* for most of the day, supports the hypothesis that the discrepancy arises not from vertical wind measurements, but from differences in the humidity sensing components.

Sensible heat flux depends on detecting the covariance between temperature and vertical wind fluctuations, while latent heat flux depends on the similar relationships between water vapor density and vertical wind. The relative errors in *H* and LE evident in this case study highlights a potential issue with the fast-response relative humidity sensor within the LI-710 rather than the companion sonic anemometer.

At UUPYF (Fig. 20b), both *H* and LE were notably higher than at UUCMF. EC150 LE peaked above 300 W m^{-2} at 15:00 MST, with LI-710 also exceeding 250 W

m^{-2}. This elevated flux activity was tightly confined to the 14:00–17:00 window, aligning with the friction velocity threshold crossing and QC grade <7 periods. Outside of that core window, both *H* and LE dropped to near-zero or negative values and many 30-minute estimates would be excluded from analysis due to poor QC Grades. While *H* and LE are often small when QC grades are > 7, this is not always the case. As shown at UUPYF, restricting analysis only to low-grade periods could eliminate substantial evaporation during such periods. Overly stringent QC filtering may overestimate mean daily/monthly fluxes (by excluding low but valid data) or underestimate total daily evaporation (by omitting brief but important events).

3.4.3 Cool Season

To further explore differences in turbulent transport between the two sites, Figs. 21-23 display wind rose plots of momentum, sensible heat, and latent heat flux, respectively, for all 30-minute periods that passed quality control (Grades 1–6). These directional distributions show important contrasts in both magnitude and source wind direction that help explain site-level differences in flux reliability, quality control, and footprint representativeness.

Momentum flux, which forms the foundation for turbulent exchange of heat and moisture, varies significantly between sites. In Fig. 21a, UUCMF exhibits a strong peak in NNW momentum transport, with roughly 20% of values exceeding 0.20 kg m^{-1} s^{-2} and a large portion falling between 0.10 and 0.20 kg m^{-1} s^{-2}. The large values found at UUCMF correspond to friction velocity values exceeding 0.3 m s^{-1}, surpassing the threshold for flux measurements and reliable footprint calculations defined by Kljun et al.

(2004). In contrast, momentum flux is significantly weaker at UUPYF (Fig. 21b). A majority of values fall below 0.04 kg m^{-1} s^{-2} – equivalent to friction velocities below 0.2 m s^{-1}. This weak turbulence has significant implications: periods with low friction velocity are more prone to non-stationarity, weak vertical mixing, and insufficient fetch, which increase the likelihood of high QC Grades. Specifically, when turbulence is weak, the assumptions of eddy covariance (e.g., well-mixed surface layer, sufficient vertical exchange) begin to break down. As a result, many low-turbulence periods at UUPYF are excluded from analysis through the quality control steps applied here, which helps to explain the substantially higher rate of poor QC grades at UUPYF compared to UUCMF.

The directional distribution of *H* further emphasizes this contrast. At UUCMF (Fig. 22a), higher sensible heat fluxes during NNW-NNE flows with the majority of 30-minute averages between 15 and 45 W m^{-2} during the cool season. UUPYF (Fig. 22b), however, shows *H* that are also substantial but more directionally varied. The highest values (>45 W m^{-2}) appear primarily under NNW winds – consistent with the predominant playa breeze towards land during the afternoon. However, some high values are also observed during westerly flow periods.

At UUCMF (Fig. 23a), latent heat fluxes are strongest from the NNW-NNE directions, with more than half of the valid values exceeding 15 W m^{-2} and a notable proportion above 45 W m^{-2}. UUPYF (Fig. 23b) shows a more scattered pattern with respect to both direction and magnitude. High LE (>45 W m^{-2}) occur across all directional flows. The most consistent and highest fluxes coincide with WNW and NNW flows, corresponding to afternoon periods when turbulence is most developed. However, the average LE at UUPYF may be biased high, because many low-magnitude flux

periods – especially overnight and in the early morning – are excluded. These periods are often flagged due to strong stability, low momentum flux and friction velocity, and non-stationary turbulence, leading to high QC Grades. While the fluxes during these periods may be small, their exclusion inflates the calculated averages. At the same time, their absence contributes to an underestimation of total daily evaporation, since LE data is missing sustained, albeit small, contributions from stable periods.

Daily observations during the entire cool season are synthesized in Fig. 24 for both sites. The number of 30-minute samples available each day after applying quality control checks are shown as well as the LI710 and EC150 daily estimates of median *H* and LE, and total evaporation.

Fig. 24a indicates the number of 30-minute observations available per day before and after quality control at UUCMF. Consistent with earlier wind rose and turbulence analyses, UUCMF retains a substantially higher number of observations per day following filtering than at UUPYF (Fig. 24e). While both sites occasionally experience instrument downtime or communication gaps, the primary cause of lower data retention at UUPYF is the higher prevalence of QC Grades > 6. This stems from frequent weak and non-stationary turbulence and stable nighttime conditions, which elevate QC flags and lead to widespread exclusion of data, particularly in the early morning and overnight hours. As a result, many low but valid flux observations are systematically removed at UUPYF.

Median daily *H* from both sensors is shown during the cool-season at UUCMF and UUPYF in Figs. 24b and 24f, respectively. UUPYF exhibits higher day-to-day variability in *H* compared to UUCMF, which is consistent with more sporadic turbulence

across the barren surface that is occasionally water covered, wet mud, or dry soil. However, the EC150 and LI-710 generally maintain good agreement in *H* at both sites. At UUCMF, the EC150 and LI-710 sensible heat fluxes are somewhat smoother, reflecting the more homogeneous flux footprint areas of the mineral ponds during the cool season.

Figs. 24c and 24g show key differences in LE at UUCMF and UUPYF, respectively. At UUCMF, EC150-based median LE is consistently higher than that from the LI-710, with the latter underestimating both the magnitude and day-to-day variability. At UUPYF, however, the sensor comparisons are closer. The LI-710 still underestimates LE relative to the EC150, but the differences are smaller and more variable than at UUCMF. This closer agreement may be misleading. In addition to known underestimation by the LI-710, the EC150 values at UUPYF may also be biased low due to the exclusion of many valid but low-magnitude LE observations during stable or weakly turbulent periods. As such, the apparent agreement between sensors at UUPYF may reflect not greater accuracy, but the fact that both systems are underestimating LE for different reasons.

Figs. 24d and 24h show total daily evaporation, summed from the 30-minute LE values that pass QC at UUCMF and UUPYF, respectively. At UUCMF, EC150-based totals are consistently higher than those from the LI-710, consistent with the underestimates of latent heat fluxes by the latter. At UUPYF, this difference persists and is even more pronounced during evaporation events in late winter and early spring. This growing discrepancy may be partially attributable to drift in the fast-response relative humidity sensor within the LI-710, exacerbated by the inability to service the sensor due

to the inaccessibility of the site when the playa surface is muddy. Sensor drift could increasingly dampen the LE signal over time, further contributing to underestimation during the latter part of the season. Despite periods of strong afternoon evaporation, daily totals at UUPYF are likely underestimated due to neglecting latent heat fluxes during stable periods. As discussed above, while average daily LE may be biased high due to the exclusion of low flux values, the total evaporation may be biased low.

Fig. 25 compares the median diurnal cycle of *H* between the LI-710 and EC150 across the entire cool-season period. Sensible heat fluxes from the EC150 and LI-710 show strong agreement throughout the day at both sites. At UUCMF, the LI-710 tracks the EC150 well but slightly underestimates the midday peak. At UUPYF, the two sensors are nearly indistinguishable for most of the day, with only minor differences during peak heating hours.

To further synthesize the results presented above, Fig. 26 compares the median diurnal cycle of LE between the LI-710 and EC150 across the entire cool-season period. At both sites, the EC150 reports higher LE values across the day, particularly during peak afternoon hours. This difference is most pronounced at UUCMF, where the LI-710 not only underestimates the midday peak but also shows a shorter duration of elevated LE, suggesting dampened sensitivity to fluctuations in water vapor. In contrast, at UUPYF, the two sensors agree more closely in shape and timing of the diurnal cycle, though the LI-710 still consistently underreports flux magnitude. At both sites, the EC150 reports consistently higher LE values than the LI-710, with the largest discrepancies during midday hours when fluxes peak. The relative difference is more pronounced at UUCMF, where the LI-710 underestimates both the magnitude and duration of the afternoon peak

compared to the EC150. The diurnal pattern is sharper and more symmetrical at UUPYF, reflecting stronger daytime evaporation.

Fig. 27 compares the monthly medians over the October 2024 – March 2025 period at both sites for *H*, LE, and evaporation rates. The differences between those monthly medians are shown as well as the daily median differences. At both UUCMF and UUPYF, *H* monthly mean differences are generally less than a few W m^{-2} with even the daily differences typically less than 10 W m^{-2}. LE and evaporation monthly median differences, however, are negative indicating the LI-710 consistently underestimates LE and evaporation rates compared to the EC150. The magnitude of both discrepancies also increases as the collection period progresses. As stated earlier, this indicates our inability to access and replace the pumps in the LI-710 sensors, sensor drift or degradation in the LI-710's fast-response relative humidity sensor, which emphasizes the need for regular maintenance and calibration to ensure long-term data reliability.

Table 2: Proportion of wind direction within 29° of sonic coordinate system.

Range	**UUCMF**	**UUPYF**
+/-10°	1.0%	2.1%
+/-29°	2.4%	6.1%

Table 3: Frequency of boundary layer conditions at UUCMF and UUPYF.
UUPYF shows higher proportion of both very unstable, stable, and very stable boundary layer conditions than UUCMF, which has a higher proportion of neutral/unstable values.

Stability	**MO Ratio**	**UUCMF**	**UUPYF**
Very stable	$\zeta \geq 0.4$	4.4%	9.2%
Stable	$0 < \zeta < 0.4$	34.3%	41.0%
Unstable	$-0.4 < \zeta \leq 0$	Percent: 50.9%	33.0%
Very unstable	$\zeta \leq -0.4$	Percent: 10.0%	16.1%

Table 4. Variables that determine QC Grade.
Three QC flags from the EC150 lead to the Grade for each 30-minute estimate of momentum and *H* and LE (Foken et al., 2012b; Campbell Scientific 2024). Defining 1 as 'best' and 9 as 'worst' follows from Foken et al. 2012b. Estimates with Grades ≤ 6 are analyzed in this study.

Grade	**Relative Non-Stationarity**	**Relative Intermittent Turbulence Characteristic**	**Wind Direction**
1 (best)	1	1 – 2	1
2	2	1 – 2	1
3	1 – 2	3 – 4	1
4	3 – 4	1 – 2	1
5	1 – 4	3 – 5	1
6 (ok)	5	5	2
7	6	6	2
8	7 – 8	7 – 8	2
9 (worst)	9	9	3

Table 5: Fraction of Grades ≥ 7 during the entire cool season for each flux variable at UUCMF and UUPYF.

Variable	**UUCMF**	**UUPYF**
Momentum Flux	26.5%	43.0%
Sensible Heat Flux *H*	40.0%	59.1%
Latent Heat Flux LE	44.2%	62.5%

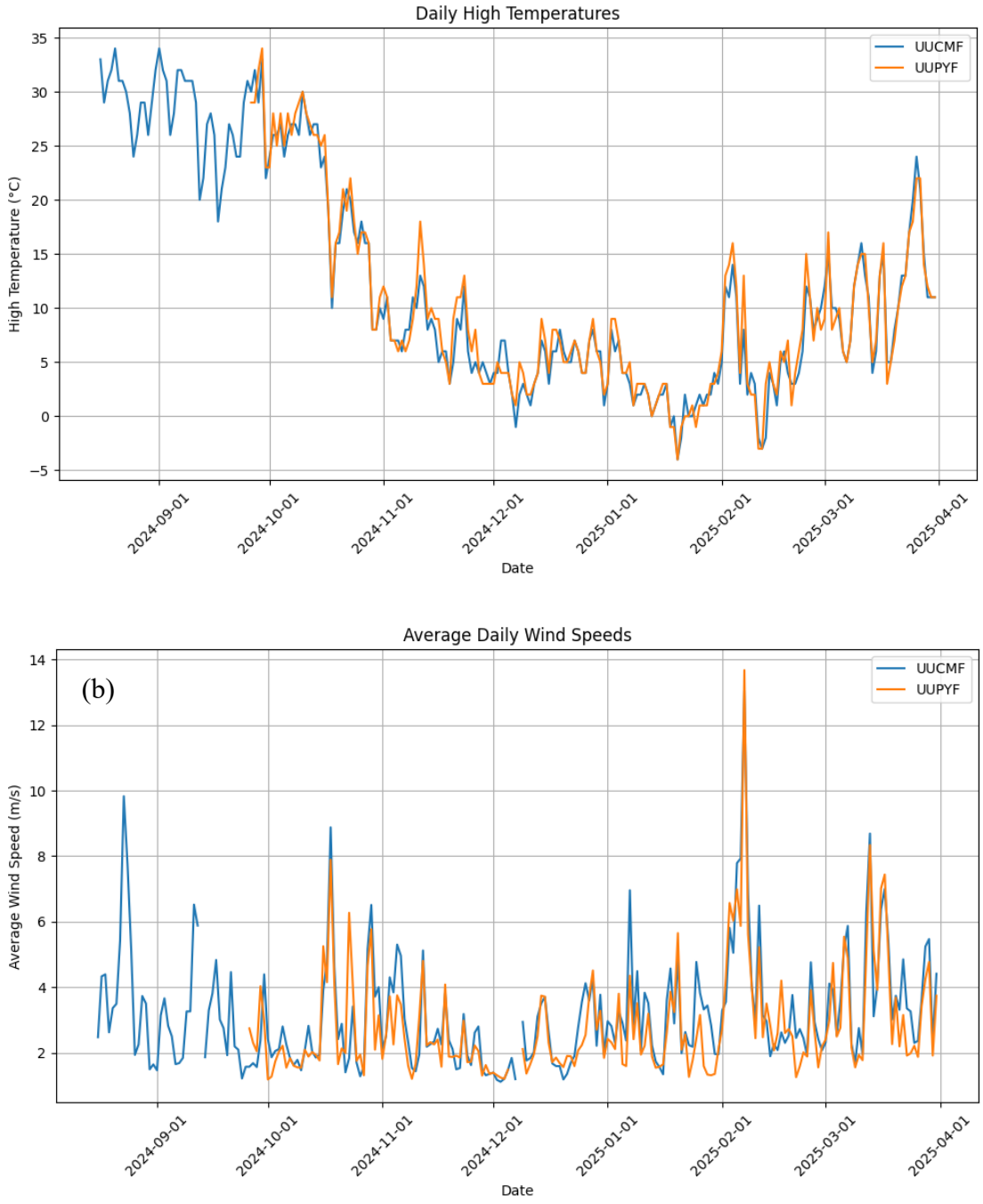


Figure 6. Daily meteorological conditions, sensor issues, and outages at the UUCMF and UUPYF stations during the data collection period. (a) Daily high temperature (°C), (b) average daily wind speeds (m s^{-1}), (c) intermittent sensor and site issues at both stations, as well as (d) partial or full site disruptions due to sensor outages.

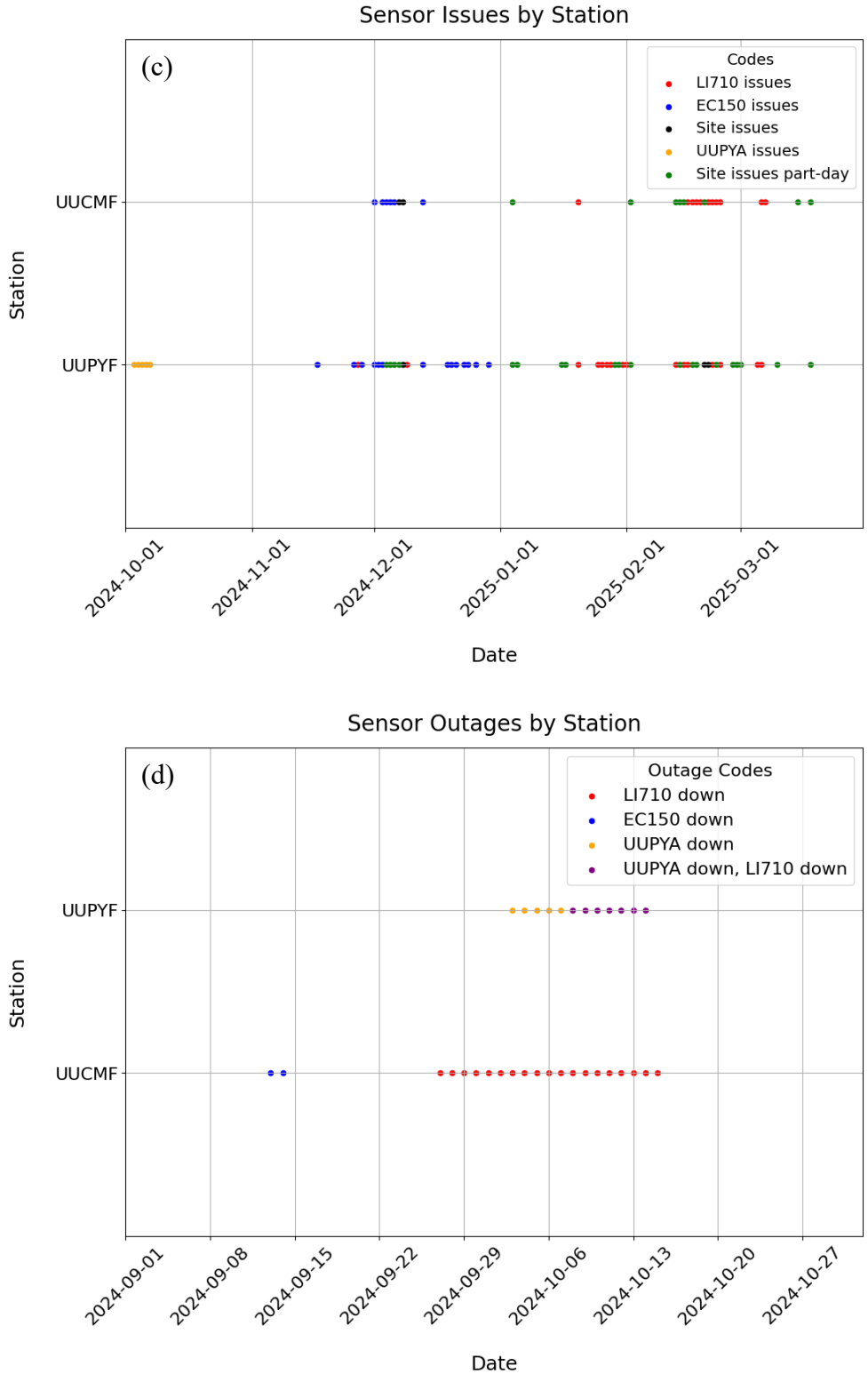


Figure 6 continued.

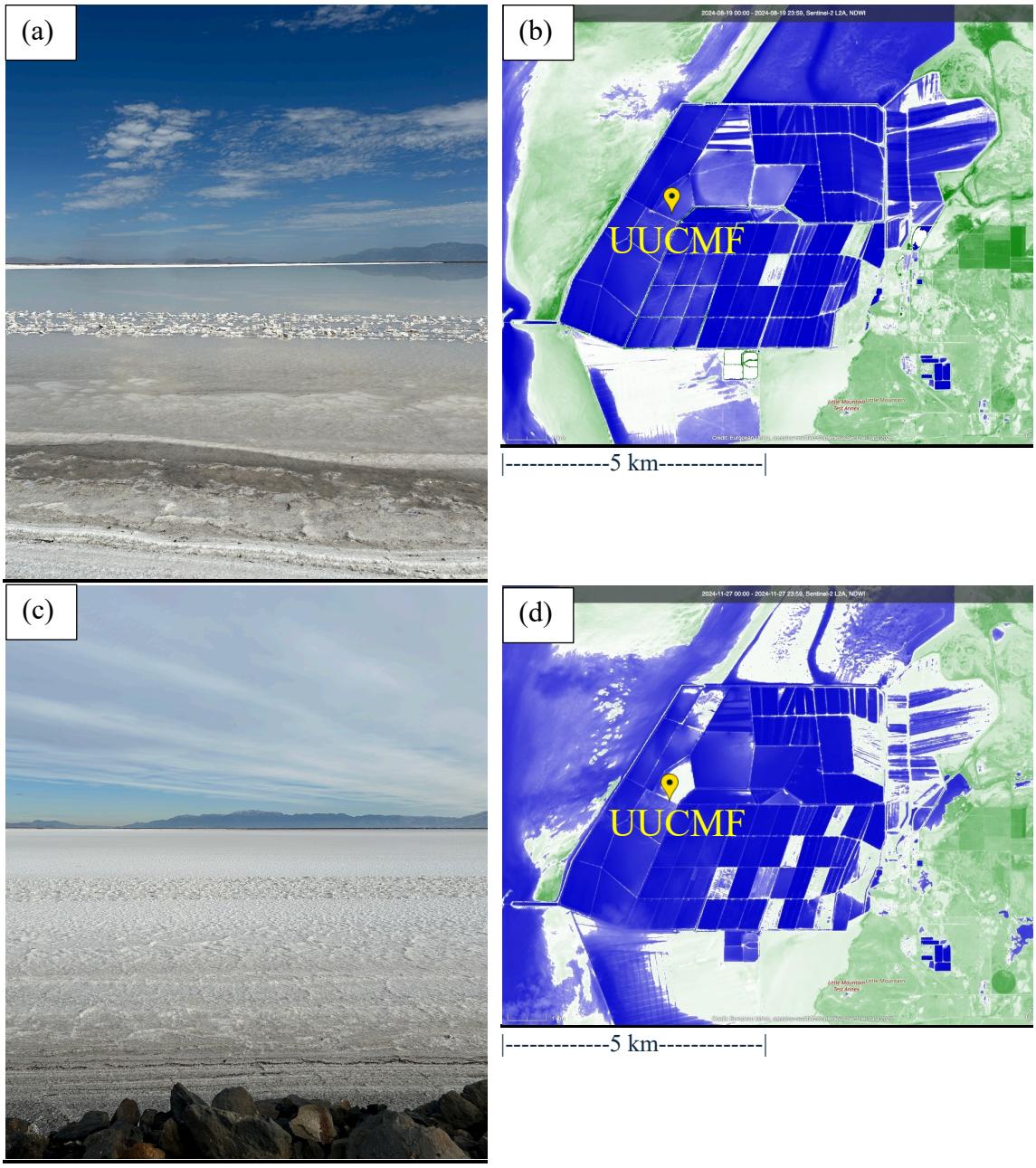
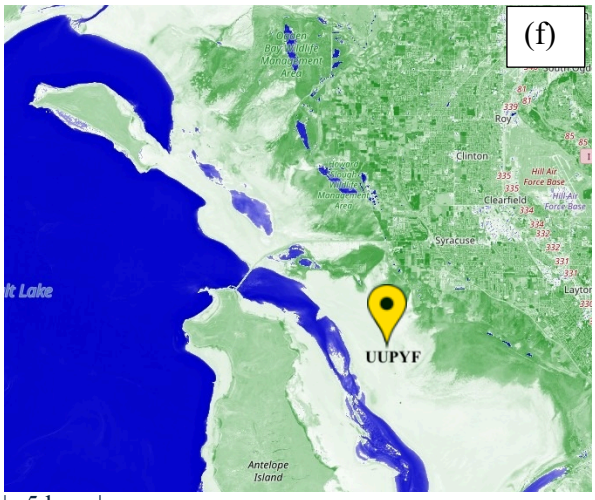


Figure 7. UUCMF and UUPYF surface conditions with corresponding Sentinel-2 NDWI images. (a) UUCMF north pond and (b) corresponding Sentinel-2 NDWI image on August 15, 2024 and (c, d) on November 24, 2024; (e) UUPYF and (f) corresponding Sentinel-2 NDWI image on September 26, 2024 and (g, h) March 19, 2024.

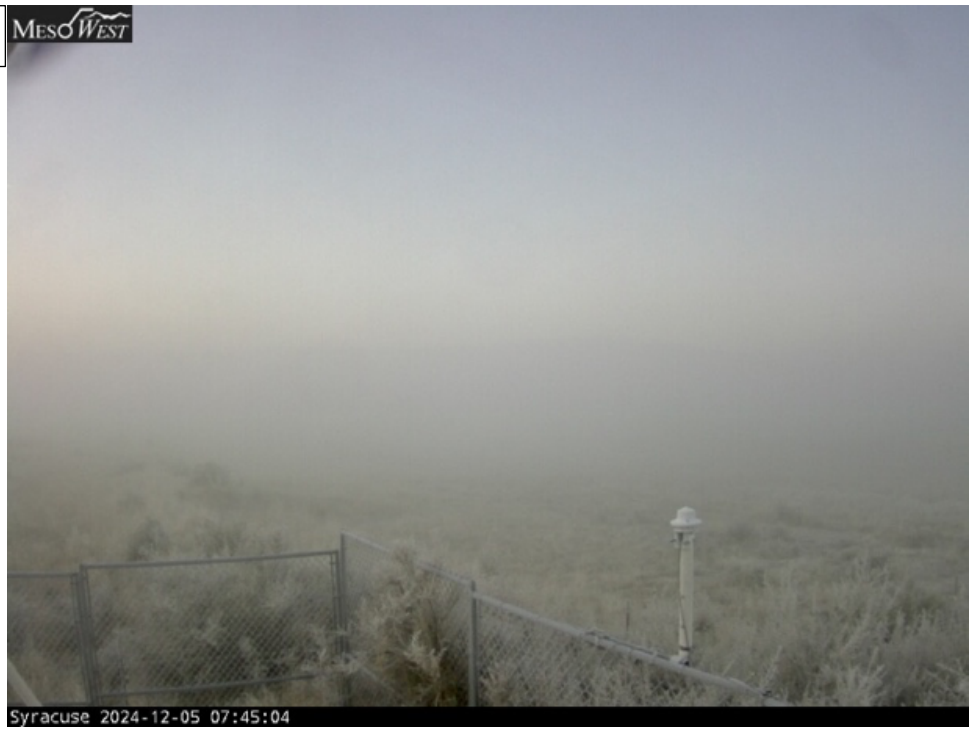


|--5 km--|

|--5 km--|

Figure 7 continued.

(a)



(b)

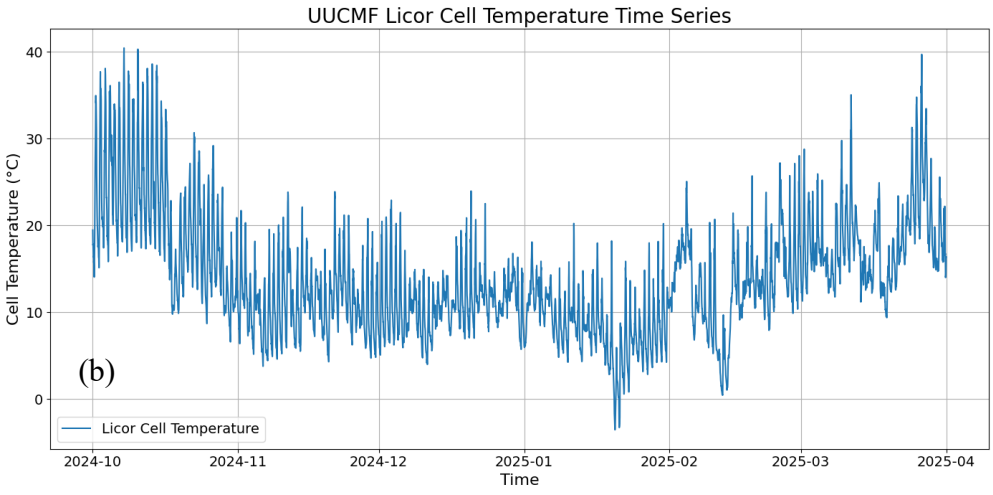
Figure 8. Conditions leading to sensor issues at UUPYF. (a) Fog at Syracuse (near UUPYF) at 7:45am MST, 5 December 2024; (b) NASA satellite image from 5 December 2024 showing fog in the same location; (c) real-time monitoring of conditions at UUPYF from MesoWest showing data issues during this period.

(c)

Tabular Listing of 51 Observations from 12/04/2024 20:00 MST to 12/05/2024 21:00 MST (ordered last to first):

Time (MST)	2.3m EC150 Temperature ° C	2.3m LI710 Temperature ° C	2.3m EC150 Dew Point ° C	2.3m LI710 Dew Point ° C	2.3m EC150 Wet bulb temperature ° C	2.3m LI710 Wet bulb temperature ° C	2.3m EC150 Relative Humidity %	2.3m LI710 Relative Humidity %	2.3m Wind Speed m/s	2.3m Wind Gust m/s	2.3m Wind Direction	EC150 Pressure mb	LI710 Pressure mb	EC150 Altimeter mb	LI710 Altimeter mb	EC150 Evapotranspiration mm	LI710 Evapotranspiration mm	Sonic Temperature ° C	Vertical Velocity m/s	Zonal Wind Standard Deviation m/s	Meridional Wind Standard Deviation m/s	Verti Wi Stand Devia m
21:00	-2.1	12.6	-7.8	-3.4	-4.1	5.3	65	32	1.9	8.1	NNW	885.23	884.00	1032.43	1031.00	0.0025	0.0000	0.6	0.00	0.90	0.26	0.
20:30	-2.5	31.1	-8.7	-3.9	-4.6	12.9	62	10	1.8	3.9	NNW	885.21	884.00	1032.41	1031.00	0.0000	0.0000	1.9	0.00	0.29	0.27	0.
20:00	-2.6	9.6		-3.9		3.7		38	2.1	4.2	NNW	885.10	883.90	1032.28	1030.88		10000005.4000	1.7	0.00	0.42	0.30	0.
19:30	-2.5	9999983.0		-594.3	-2.8	9999983.0		10000000	2.6	5.7	NNW	884.81	883.60	1031.94	1030.53		10000005.4000	1.8	0.00	0.33	0.27	0.
19:00	-2.4	28.0		-3.5	-2.5	11.8		13	1.9	2.7	N	884.74	883.50	1031.86	1030.42		0.0000	1.8	0.00	0.21	0.27	0.
18:30	-1.8	0.6		-2.9	-2.0	-0.8		78	2.2	5.1	N	884.62	883.30	1031.72	1030.18		0.0000	2.3	0.00	0.35	0.29	0.
18:00	-1.3	-0.3		-2.4	-1.3	-1.2		86	2.2	2.9	N	884.56	883.30	1031.65	1030.18		0.0000	1.8	0.00	0.23	0.27	0.
17:30	-0.4	0.6	-21.8	-1.4	-5.6	-0.2	18	86	2.8	4.2	NNW	884.62	883.30	1031.72	1030.18	0.0000	0.0000	3.7	0.00	0.35	0.50	0.
17:00	0.3	1.2	-6.3	-0.6	-2.1	0.4	61	88	3.2	4.4	NW	884.67	883.30	1031.78	1030.18	0.0000	0.0000	2.7	0.00	0.40	0.32	0.1
16:30	1.3	2.0	-1.0	0.6	0.3	1.4	85	90	3.6	4.8	NW	884.74	883.20	1031.86	1030.07	0.0000	0.0000	2.0	0.00	0.35	0.48	0.1
16:00	3.1	3.6	0.6	1.4	2.0	2.6	83	85	3.5	4.9	WNW	884.78	883.10	1031.91	1029.95	0.0178	0.0152	3.7	0.00	0.59	0.34	0.1

Figure 8 continued.



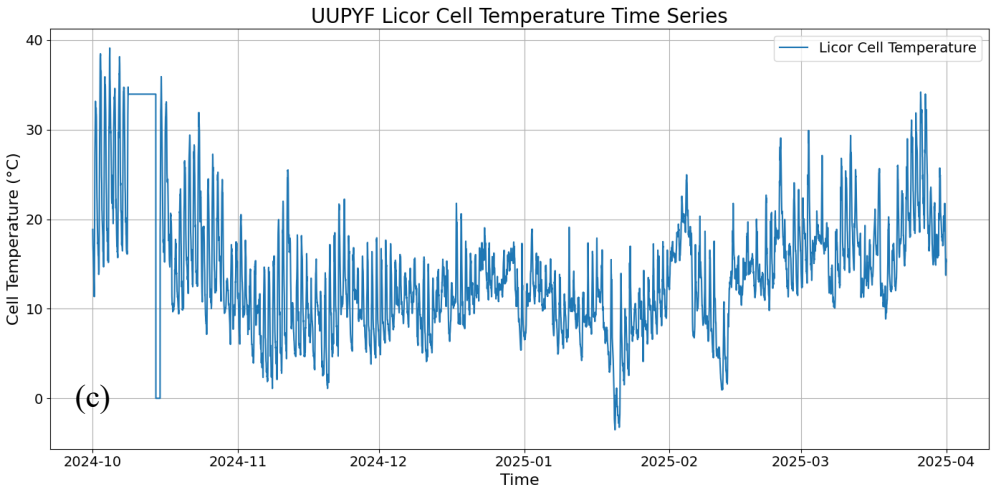


Figure 9. LI-710 cell temperature before and after insulating with pipe wrap in November. (a) LI-710 insulated with pipe wrap; LI-710 cell temperature (°C) during the cool season for (b) UUCMF and (c) UUPYF.

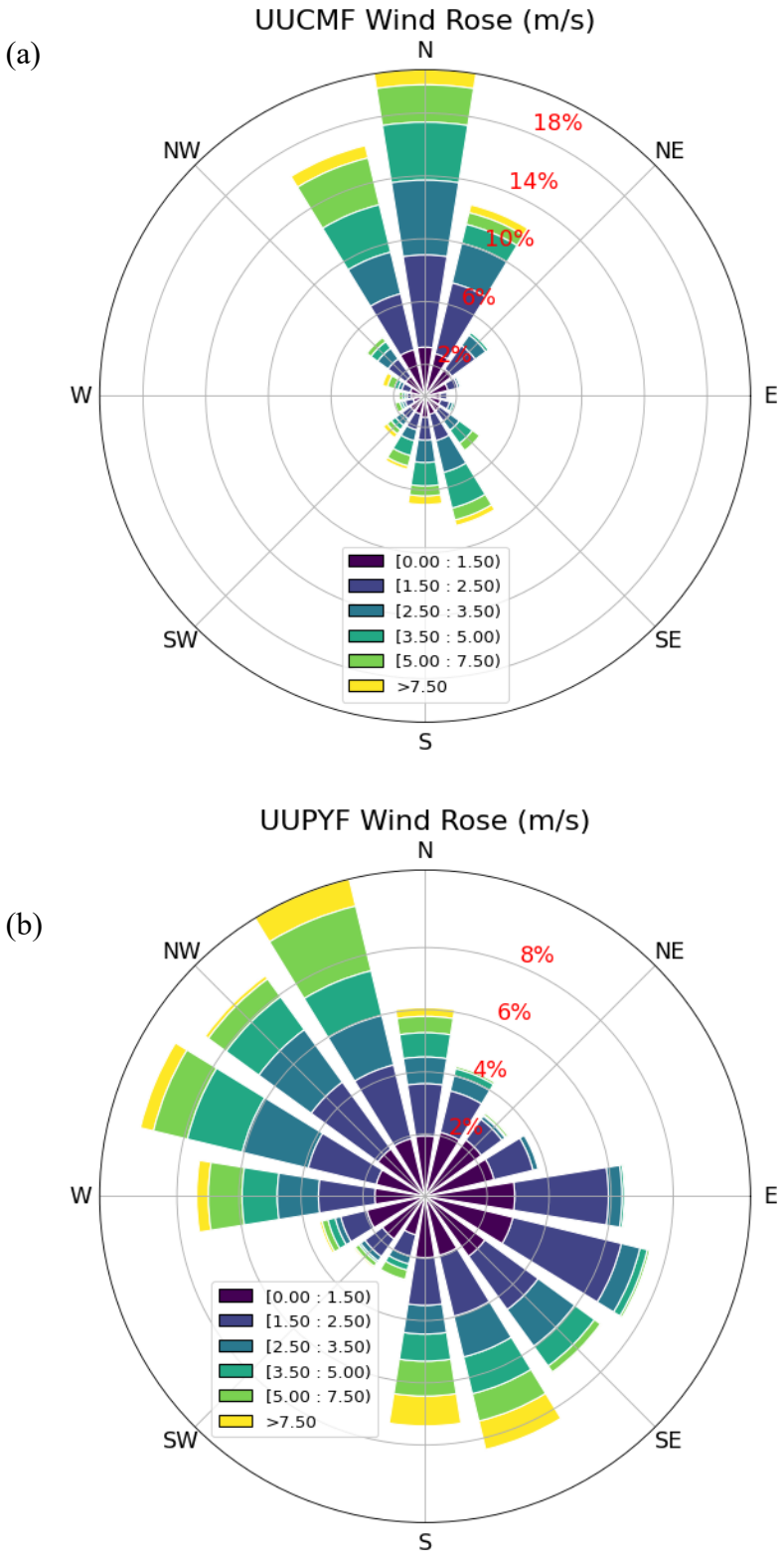


Figure 10. October 2024-March 2025 wind roses (m s^{-1}) showing proportion and intensity of winds from 16 cardinal directions. (a) UUCMF wind rose and (b) UUPYF wind rose.

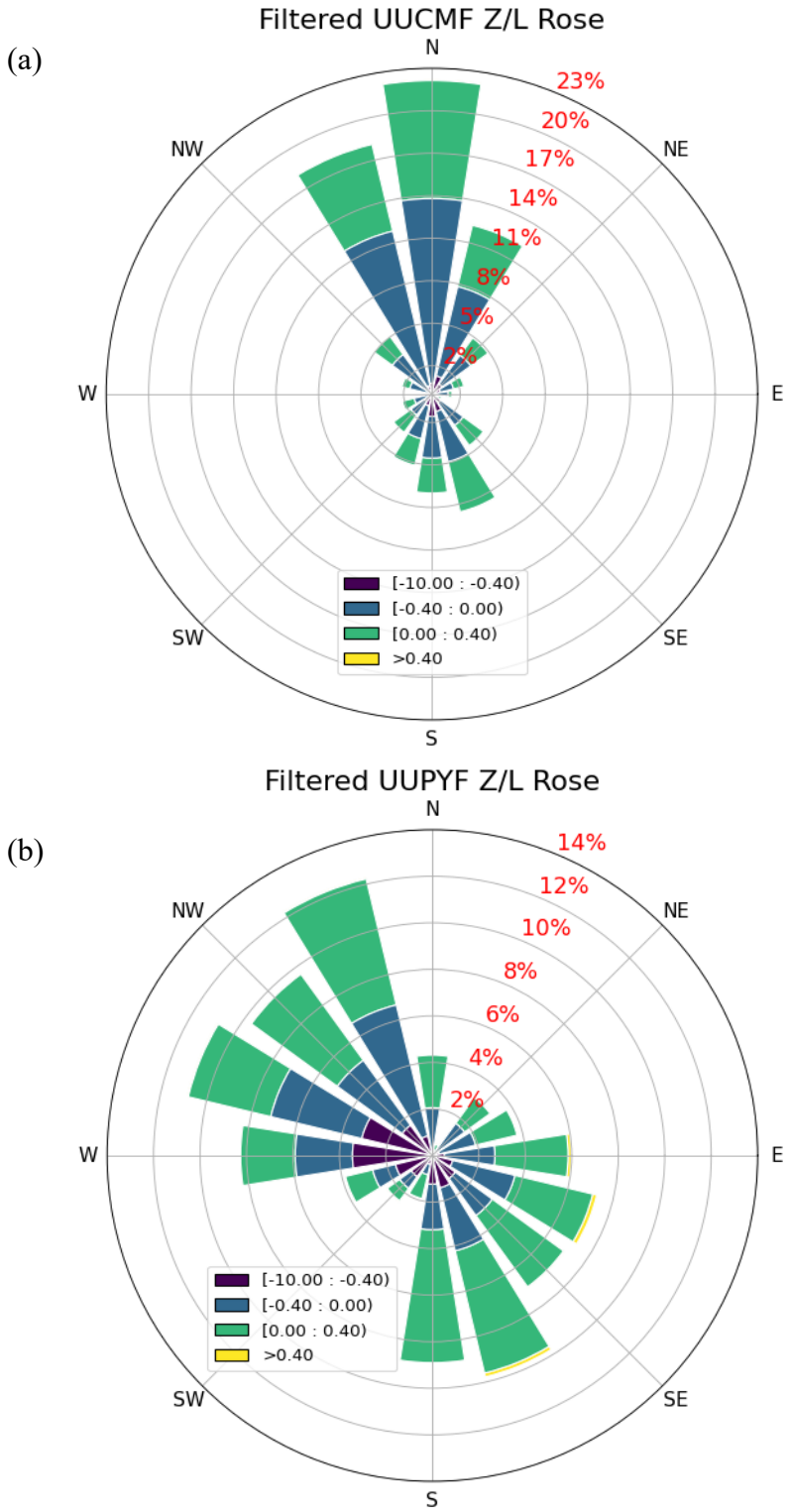


Figure 11. Distribution of boundary layer stability conditions at UUCMF and UUPYF during the cool season (October–March). Wind rose plots for (a) UUCMF and (b) UUPYF show the directional frequency of *Z/L* values.

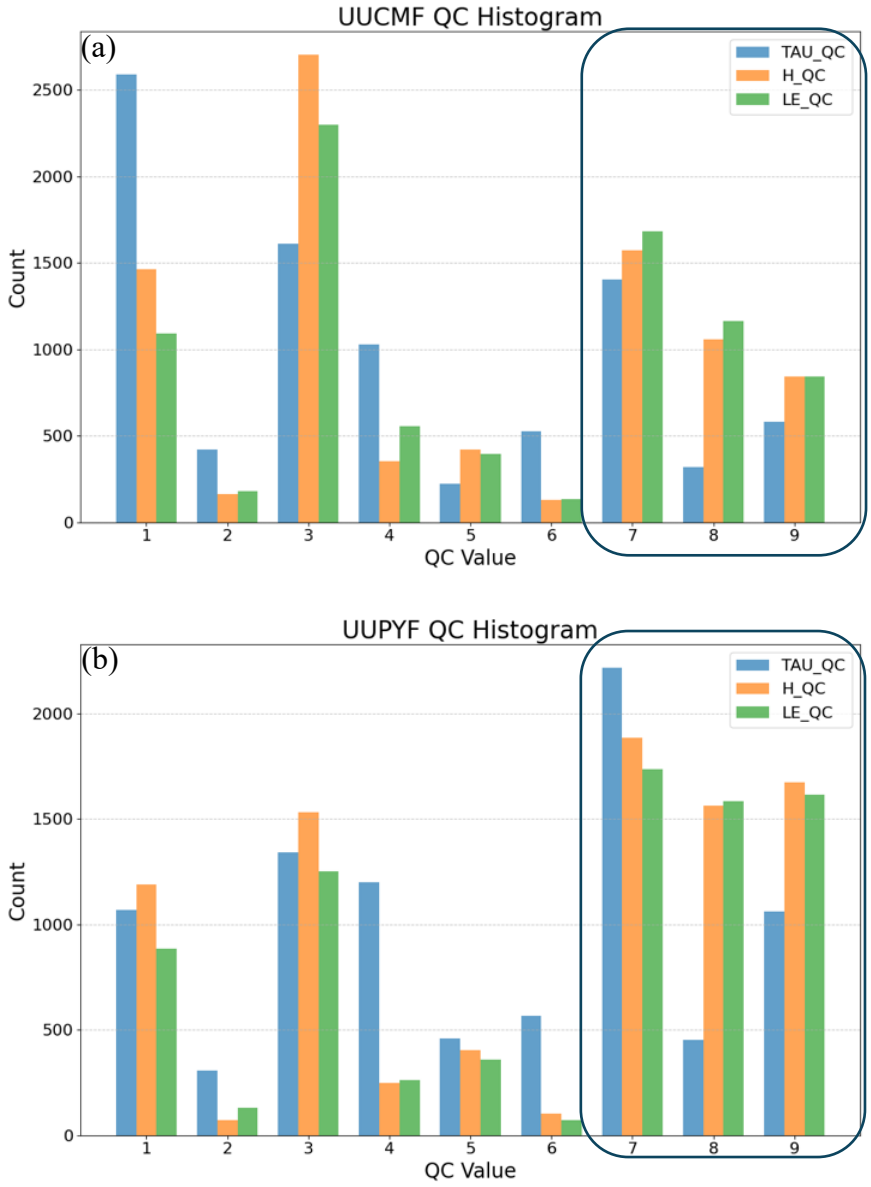


Figure 12. Distribution of EC150 Grade counts for each variable (TAU_QC: momentum, H_QC: sensible heat flux, and LE_QC: latent heat flux) during the entire cool season. (a) QC distributions at UUCMF and (b) as in (a) for UUPYF.

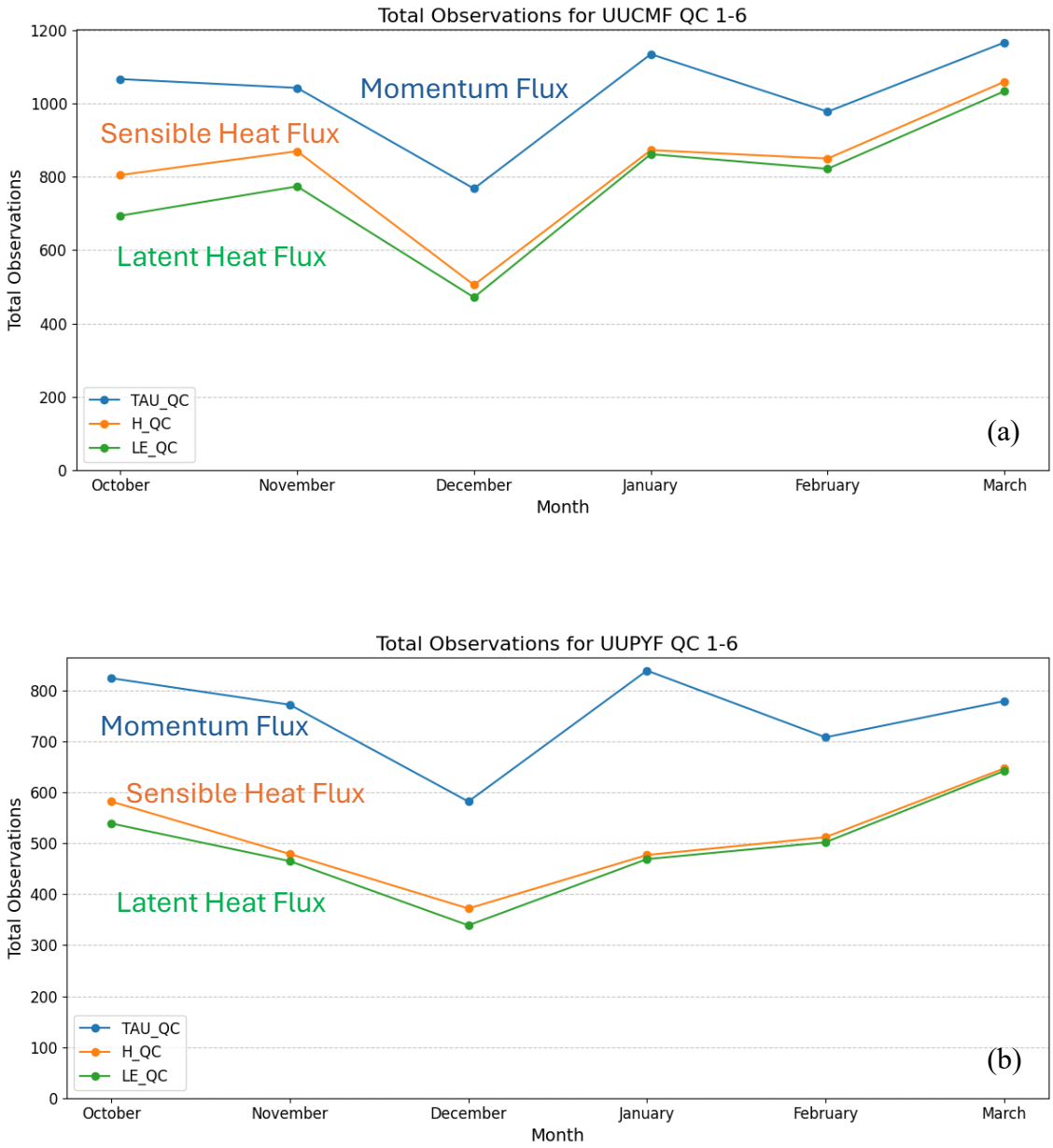


Figure 13. Monthly QC grade counts ≤ 6 for each of three variables: momentum (TAU), sensible (*H*), and latent heat flux (LE). (a) UUCMF and (b) UUPYF show different ranges in y-axis, with fewer accepted values at UUPYF than UUCMF.

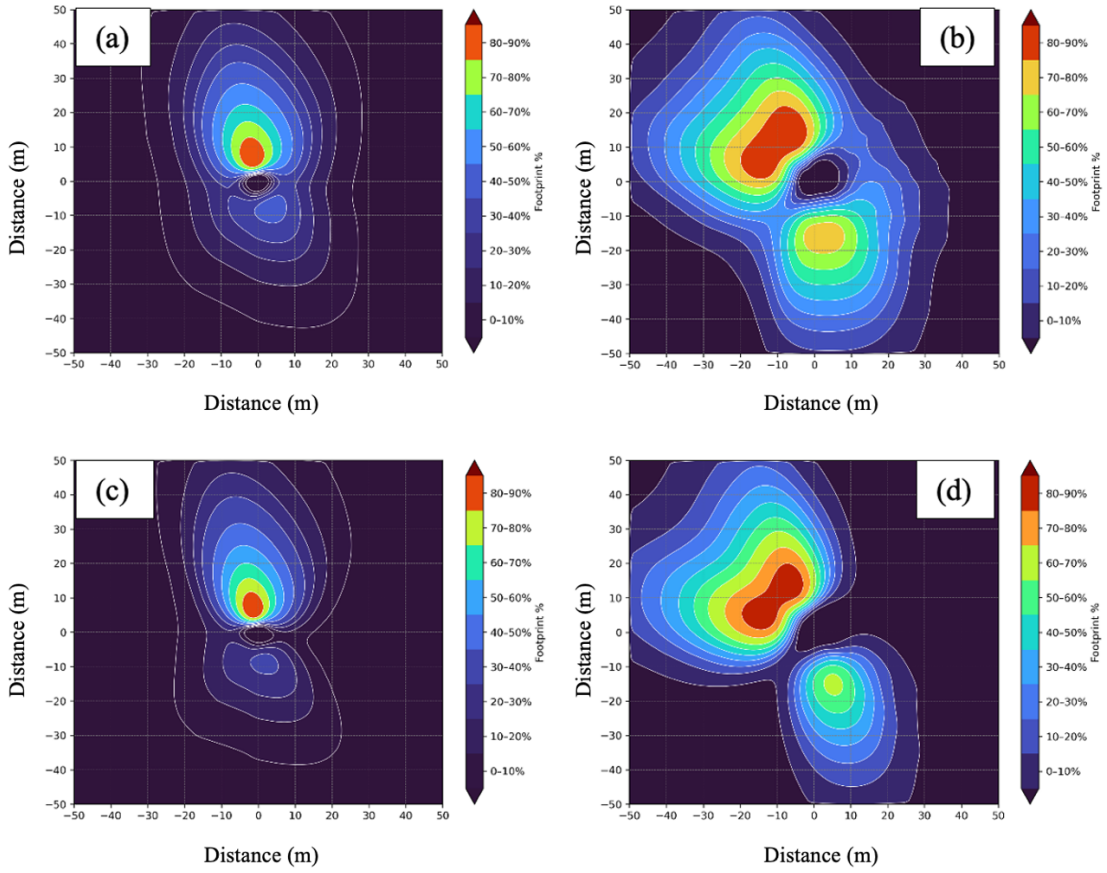


Figure 14. Footprint analysis during the cool season following Kljun et al. (2004) where higher percentages defined by the scales denote distances (m) contributing larger fractions of the fluxes. a) flux footprint at UUCMF using all samples during the cool season. b) As in (a) except for UUPYF. c) As in (a) except using 30-minute samples with grades of 6 or less. d) As in (c) except at UUPYF.

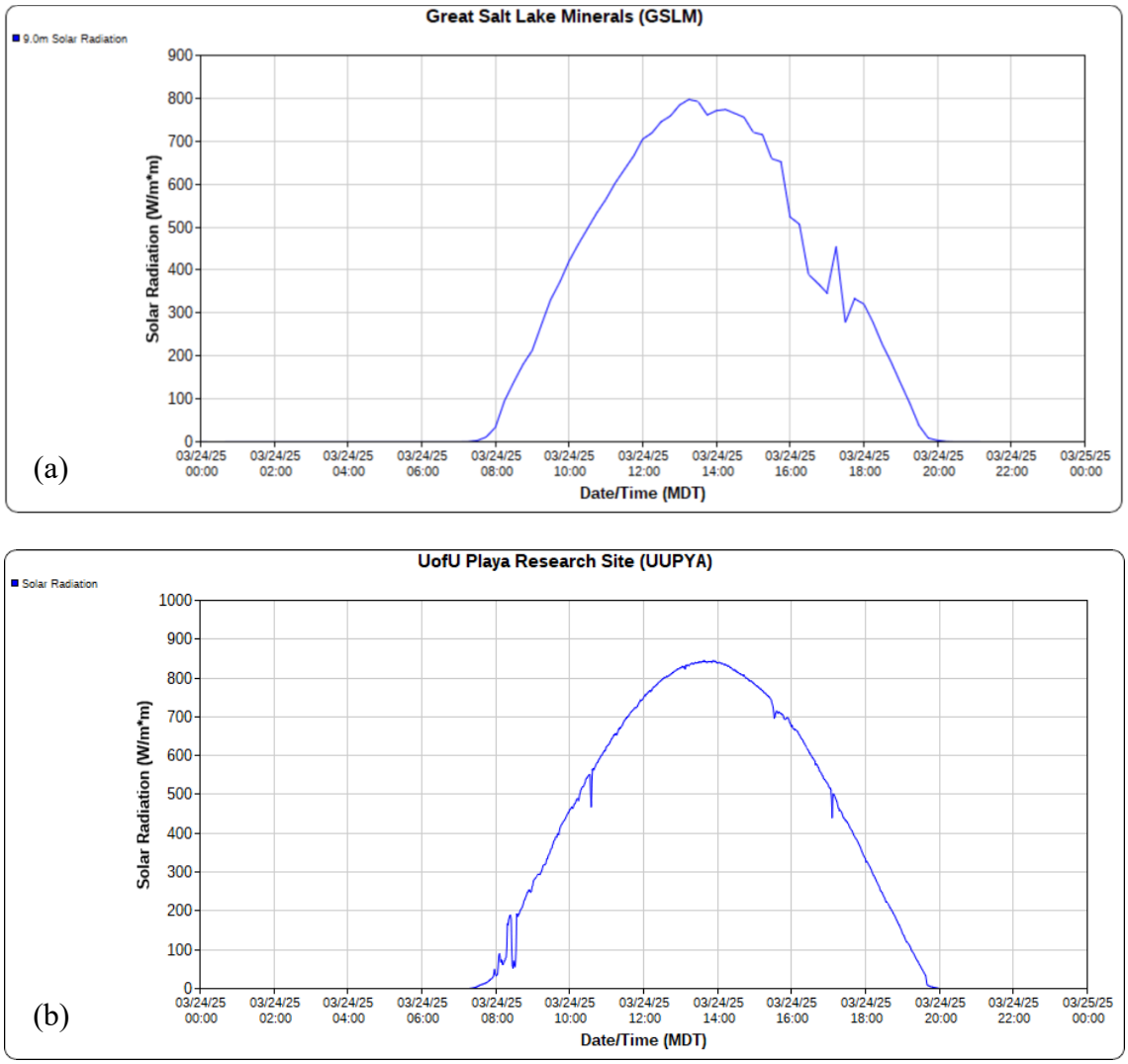


Figure 15. Solar radiation (W m^{-2}) on 24 March 2025. (a) Mineral ponds meteorological site (GSLM) and (b) playa meteorological site (UUPYA).

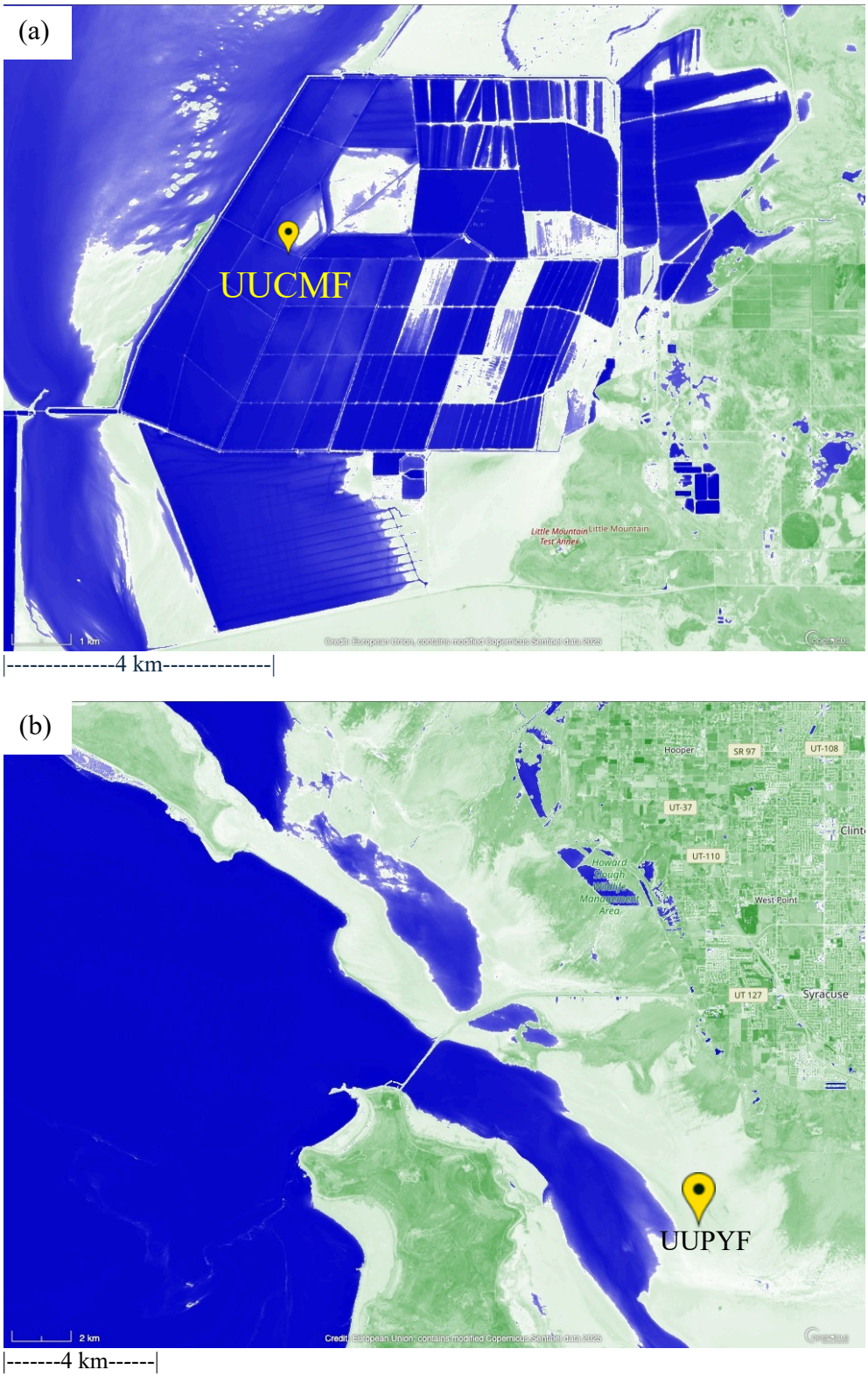


Figure 16. 24 March 2025 NDWI Sentinel-2 satellite images. a) UUCMF and b) UUPYF.

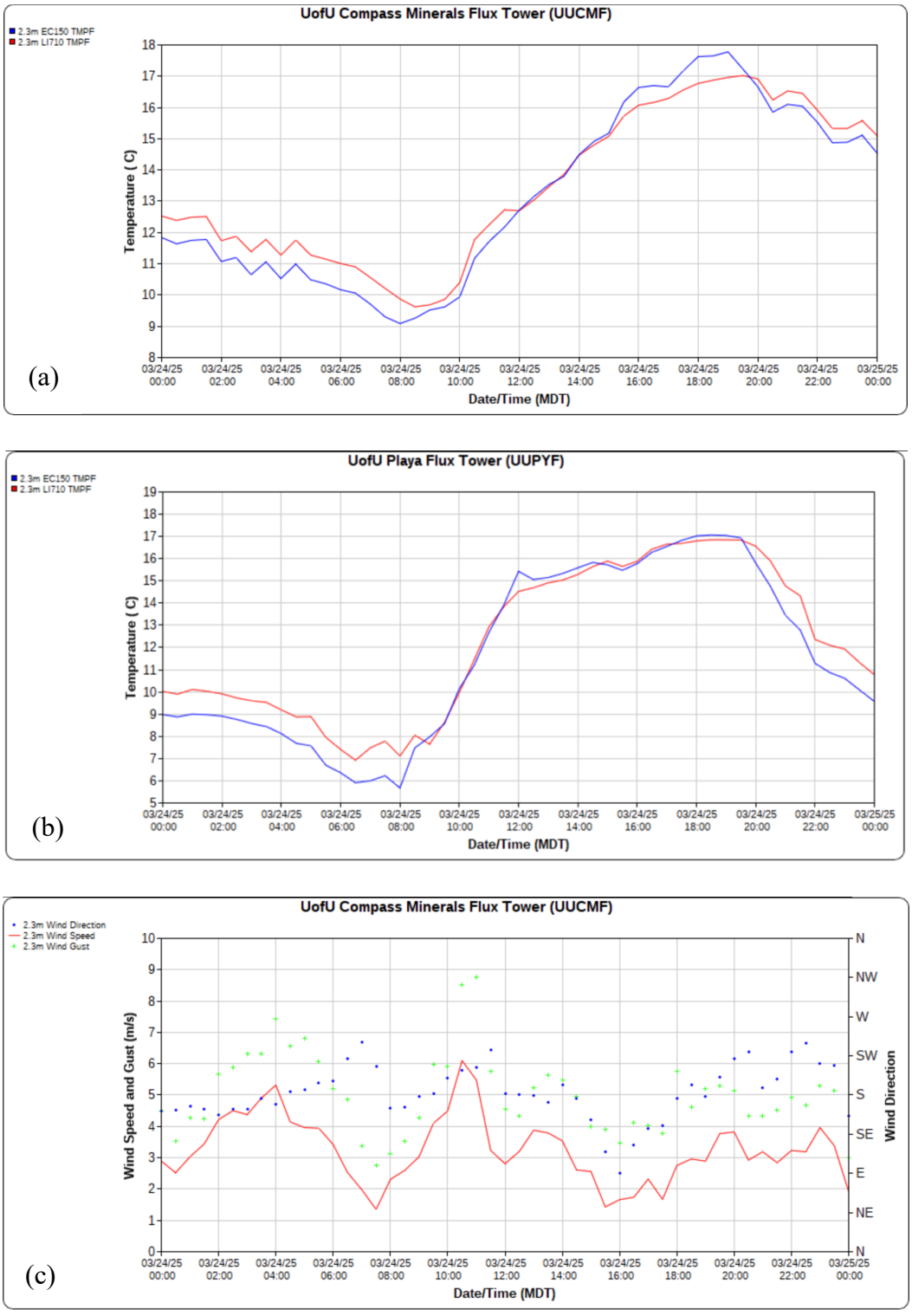


Figure 17. Temperature and wind speed at the flux sites on 24 March 2025. Diurnal cycles of 2.3 m air temperature and wind conditions at the UUCMF (a, c) and UUPYF (b, d) flux towers on 24 March 2025. Temperature plots compare EC150 and LI-710 measurements. Wind plots show speed (red line), gusts (green points), and direction (blue points) at each site.

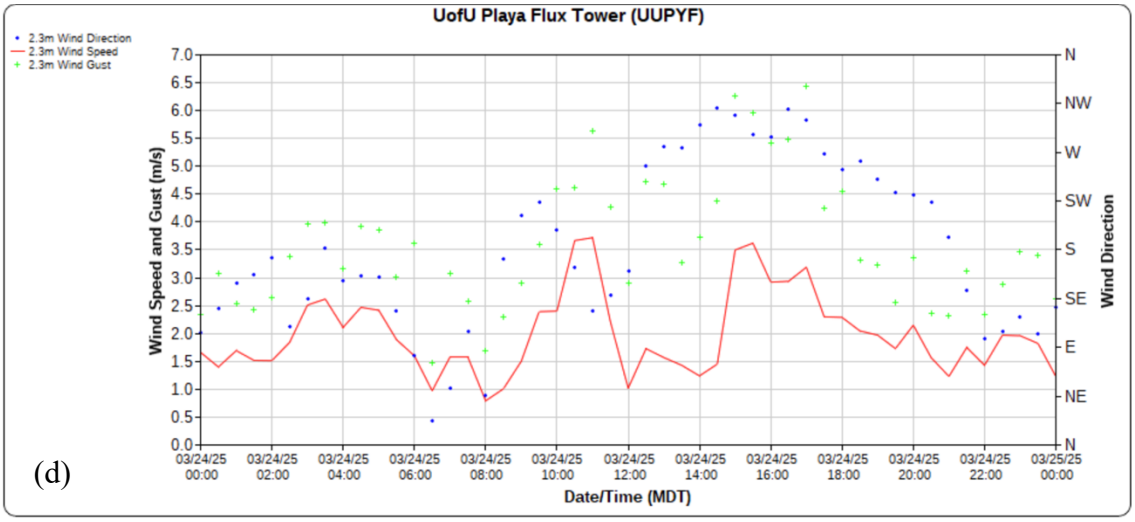
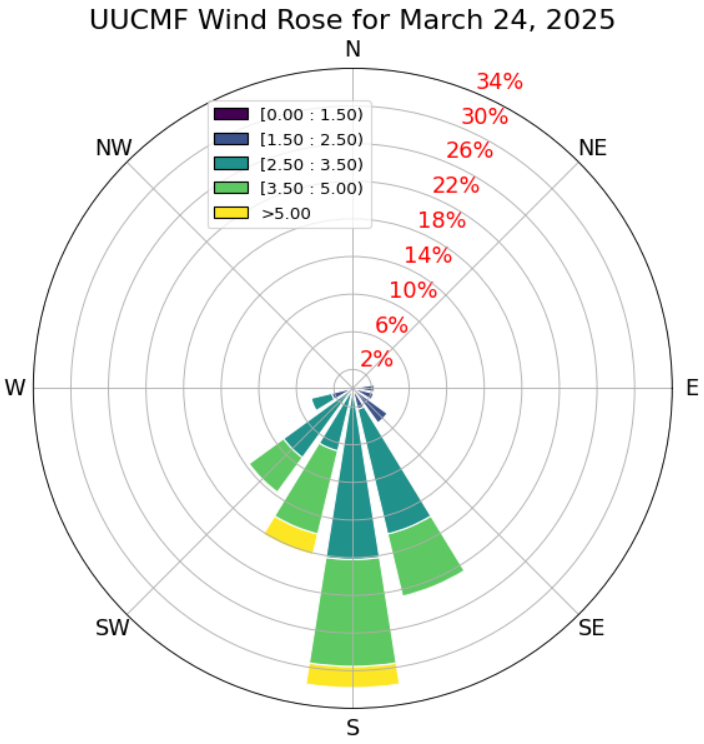


Figure 17 continued.

(a)



(b)

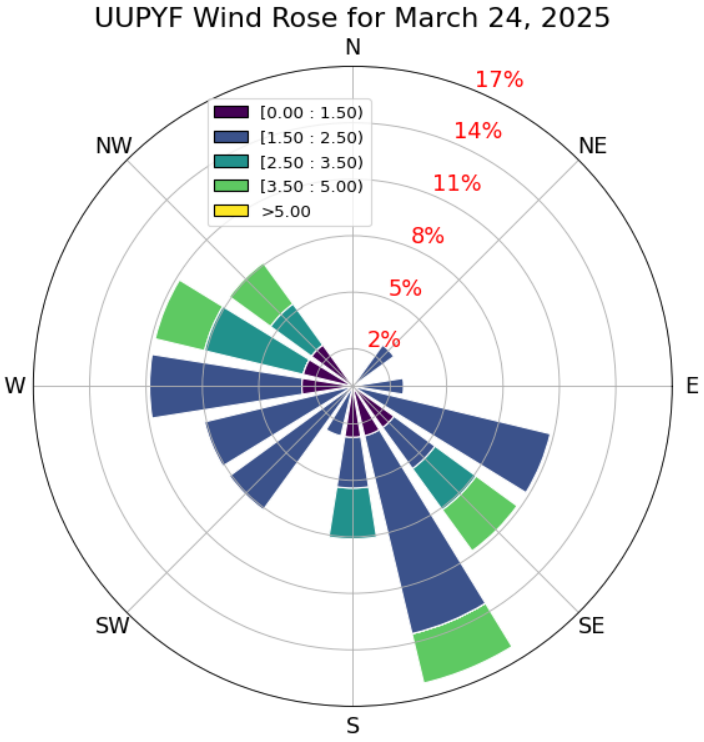


Figure 18. Wind rose for 24 March 2025. (a) UUCMF and (b) UUPYF.

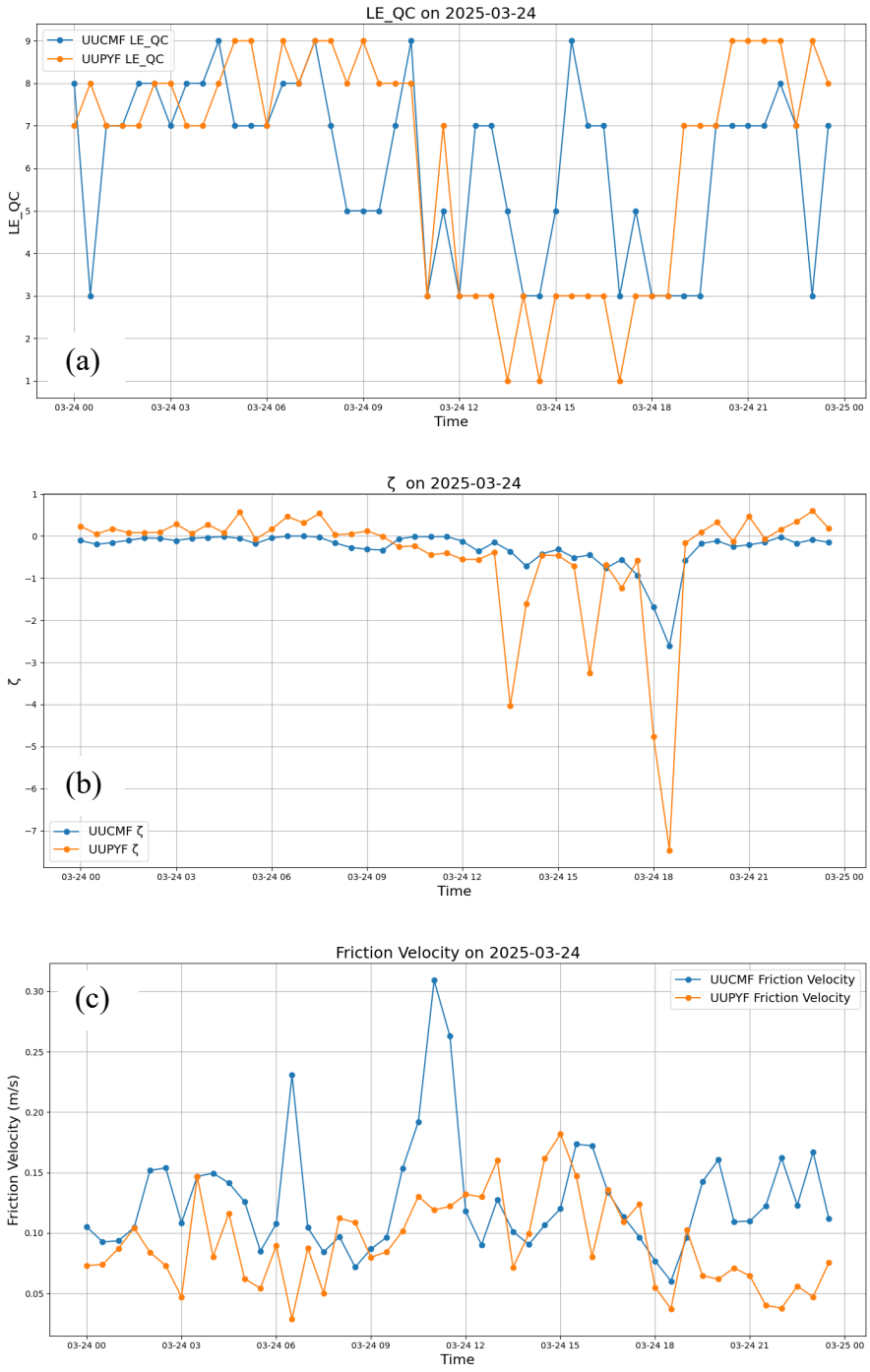


Figure 19. 30-minute time series (MDT) for important QC and meteorological variables on 24 March 2025. (a) 30-minute LE Grades; (b) 30-minute Monin-Obukhov ratio averages; (c) 30-minute averaged friction velocity (m s^{-1}).

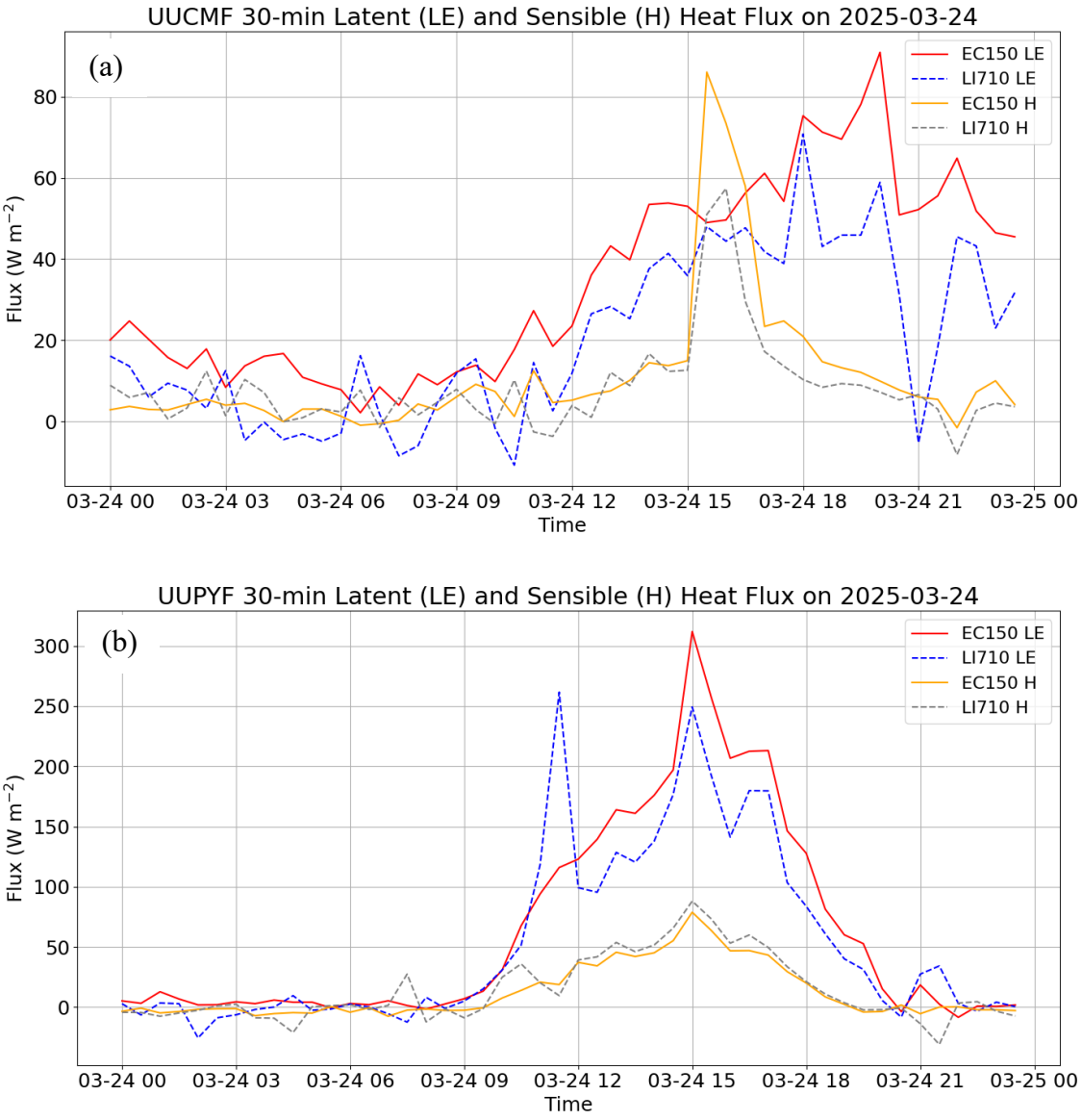
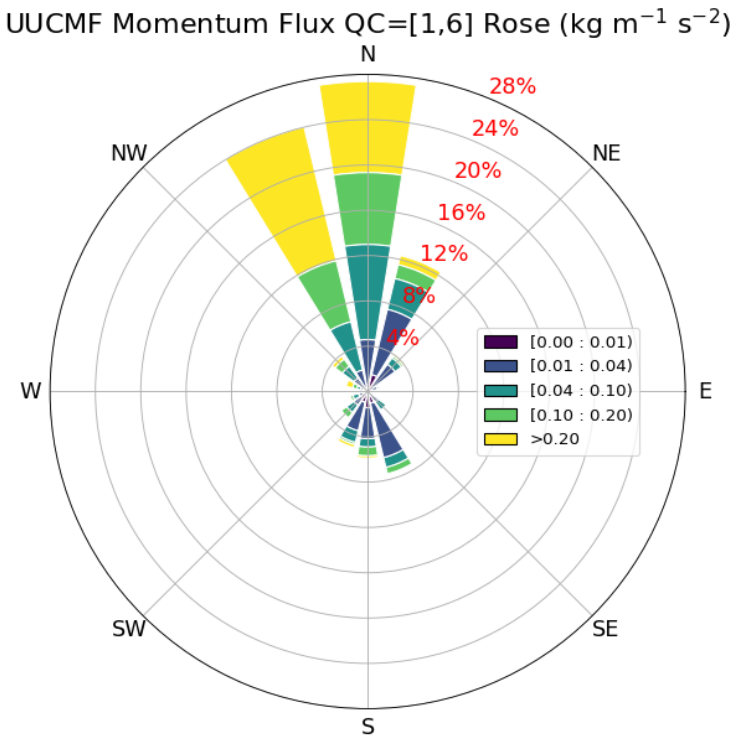


Figure 20. Comparison of 30-minute average *H* and LE (W m^{-2}) between the LI-710 and EC150 on 24 March 2025. (a) comparisons at UUCMF and (b) as in (a) for UUPYF.

(a)



(b)

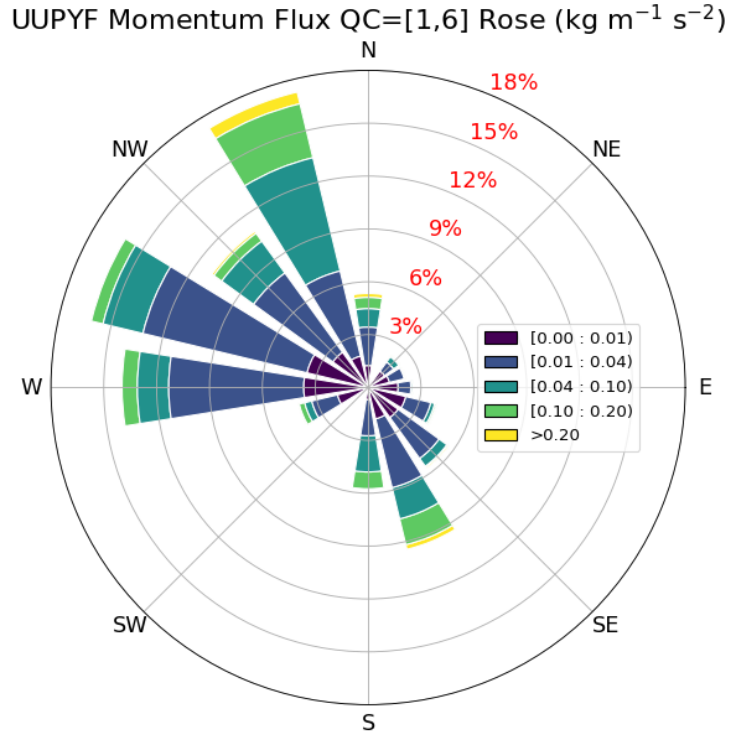
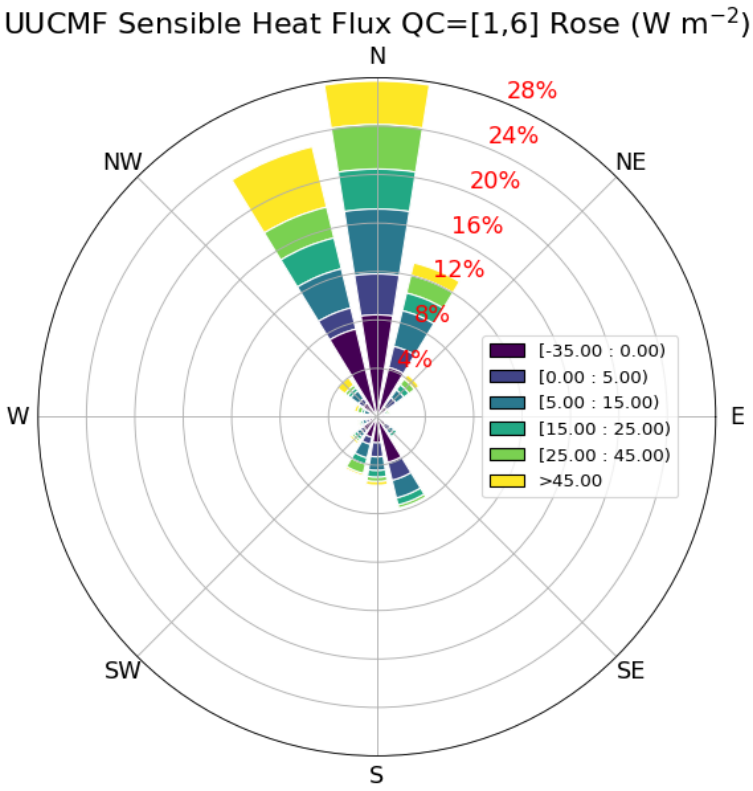


Figure 21. Directional variability in momentum fluxes (kg m^{-1} s^{-2}) during the cool season. These include periods only when QC Grades are ≤ 6 at: (a) UUCMF and (b) UUPYF.

(a)



(b)

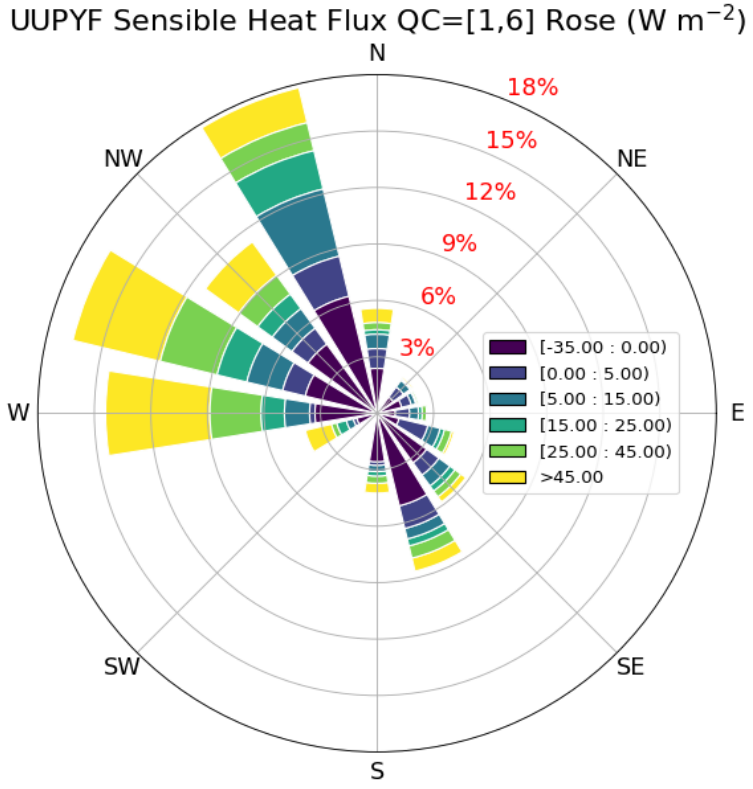
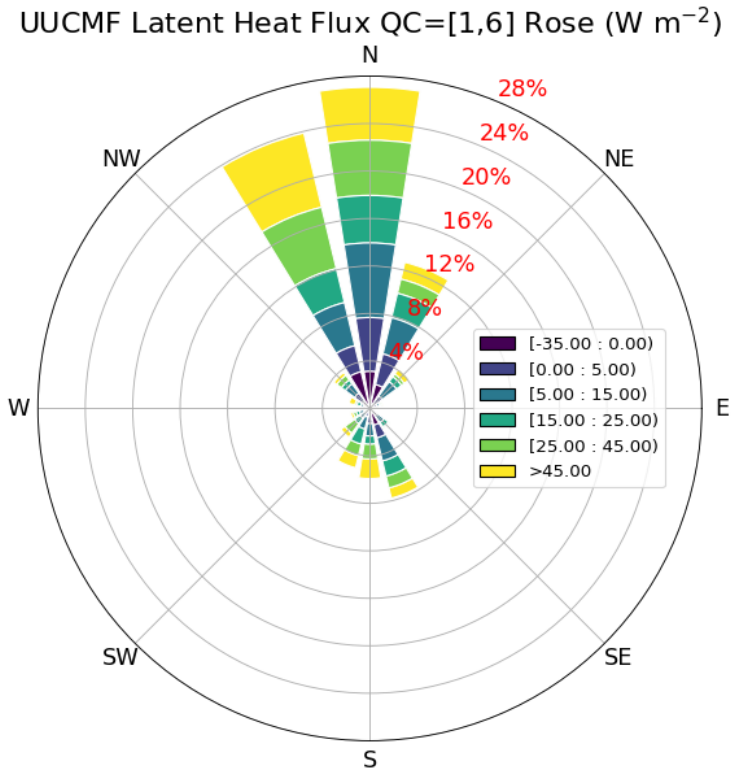


Figure 22. Directional variability in H (W m^{-2}) during the cool season. These include periods only when QC Grades are ≤ 6 at: (a) UUCMF and (b) UUPYF.

(a)



(b)

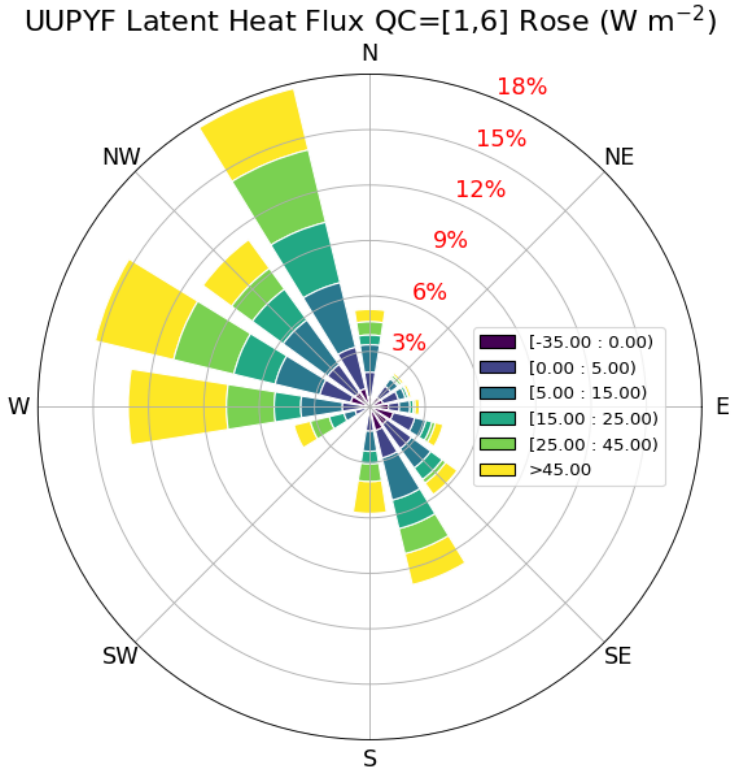


Figure 23. Directional variability in LE (W m^{-2}) during the cool season. These include periods only when QC Grades are ≤ 6 at: (a) UUCMF and (b) UUPYF.

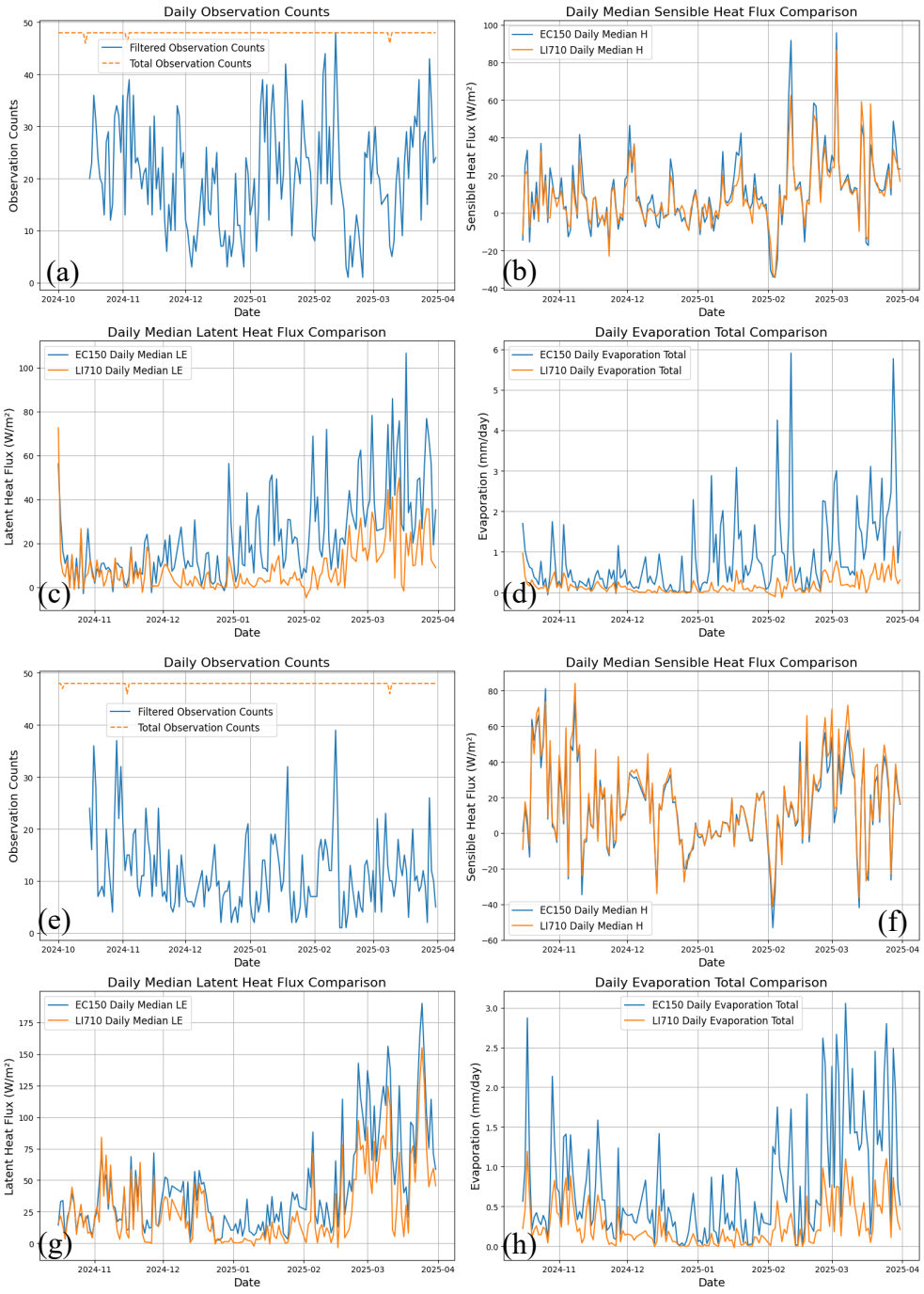


Figure 24. Flux comparisons over the entire cool season at each site. (a) Number of samples available after QC (blue) and maximum possible samples (orange) at UUCMF, (b) median 30-minute *H* (W m^{-2}) at UUCMF, (c) median 30-minute LE (W m^{-2}) at UUCMF , (d) total evaporation (mm day^{-1}) at UUCMF, (e) as in (a) except for UUPYF, (f) as in (b) except for UUPYF, (g) as in (c) except for UUPYF, (h) as in (d) except for UUPYF.

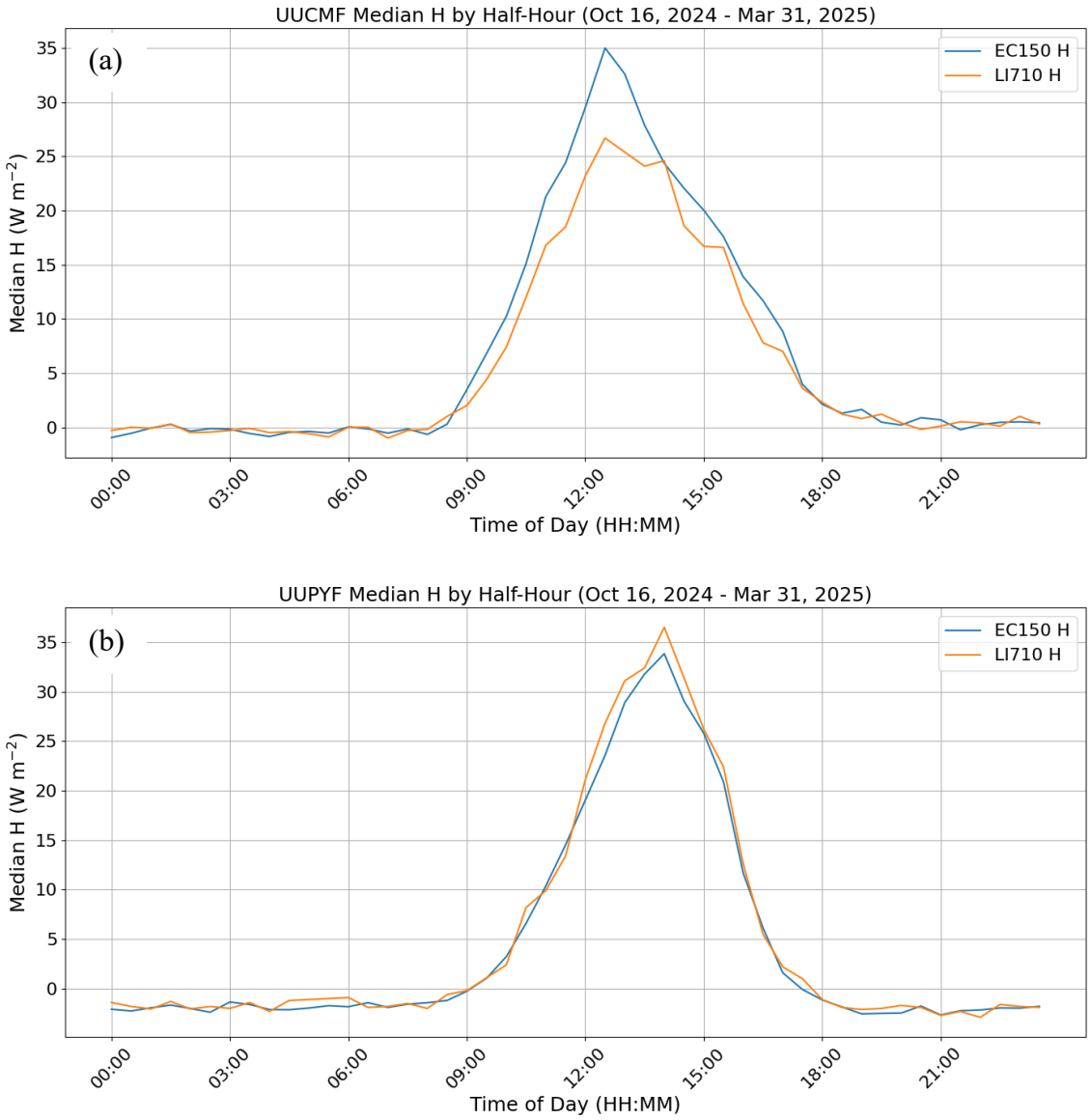


Figure 25. Median non-QC *H* by half-hour for the period 16 October 2024 to 31 March 2025. EC150 and LI-710 measurements are compared at (a) UUCMF and (b) UUPYF.

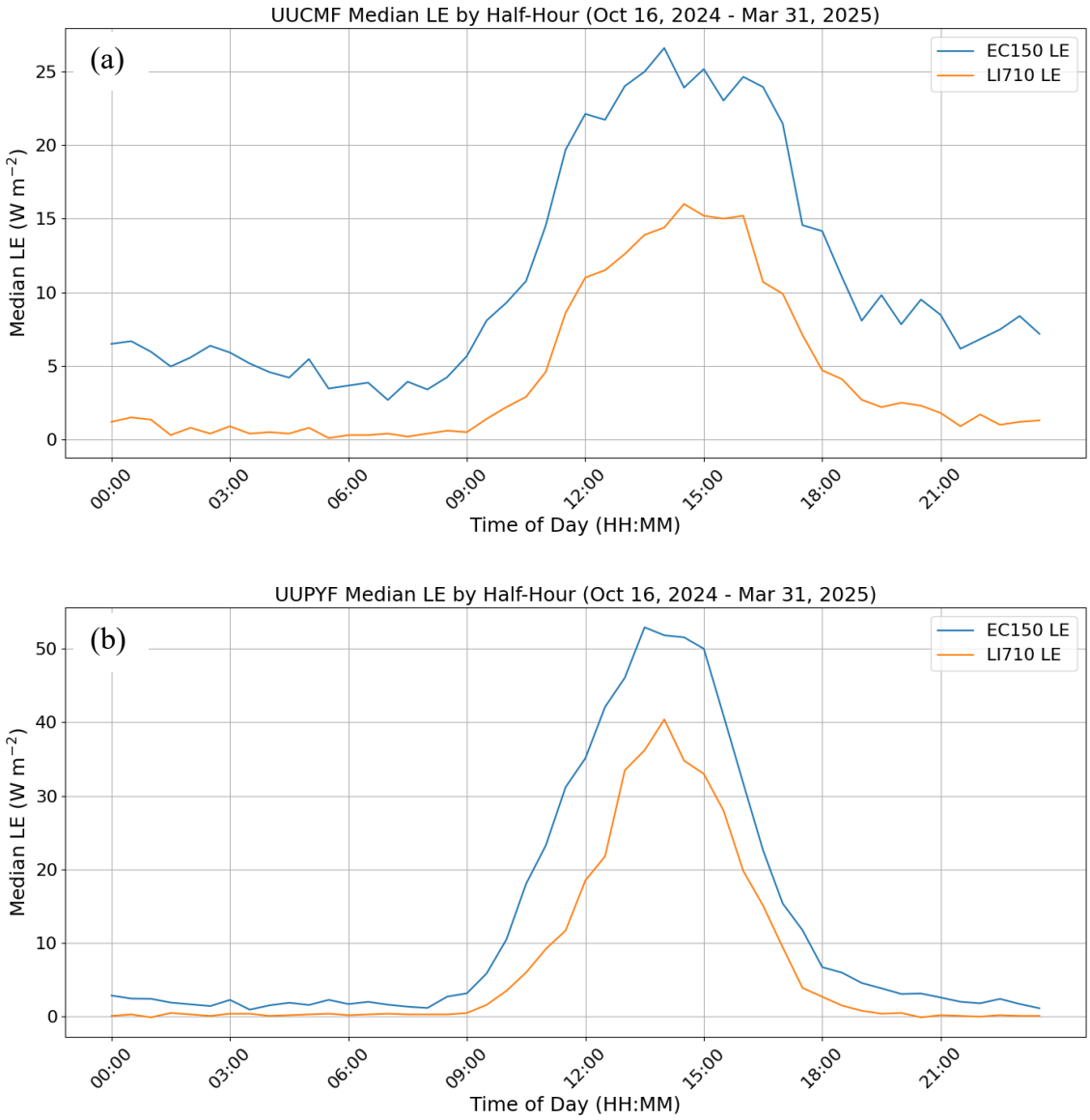


Figure 26. Median non-QC LE (W m^{-2}) by half-hour for the period 16 October 2024 to 31 March 2025. EC150 and LI-710 measurements are compared at (a) UUCMF and (b) UUPYF.

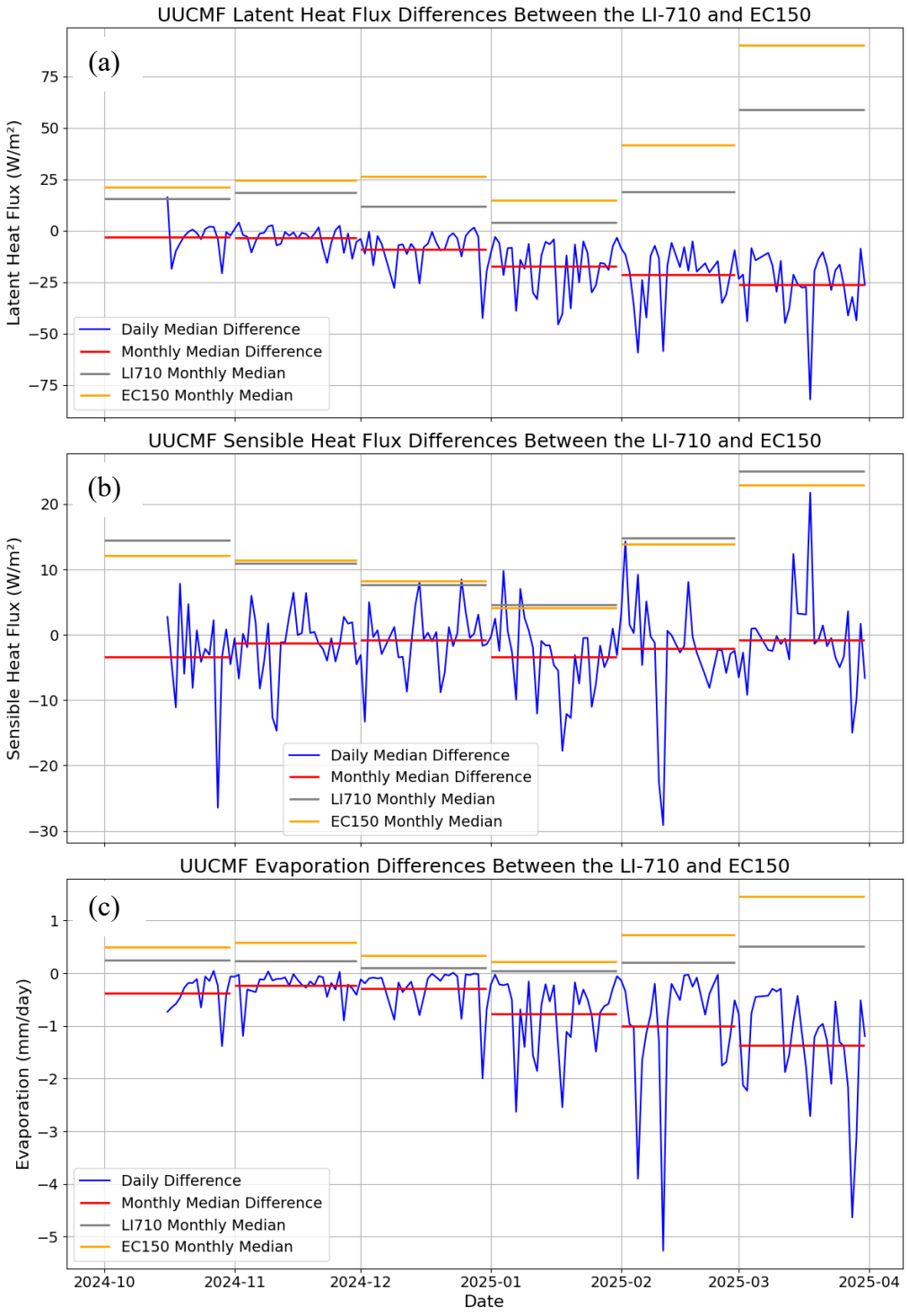


Figure 27. Daily and monthly differences and medians in flux measurements comparing the LI-710 and EC150. (a) Monthly medians at UUCMF of LE from LI-710 (gray) and EC150 (orange) and monthly and daily differences between the LI-710 and EC150 medians (red and blue, respectively). (b) As in (a) except for H. (c) as in (a) except for evaporation, (d) as in (a) except for UUPYF, (e) as in (b) except for UUPYF, (f) as in (c) except for UUPYF.

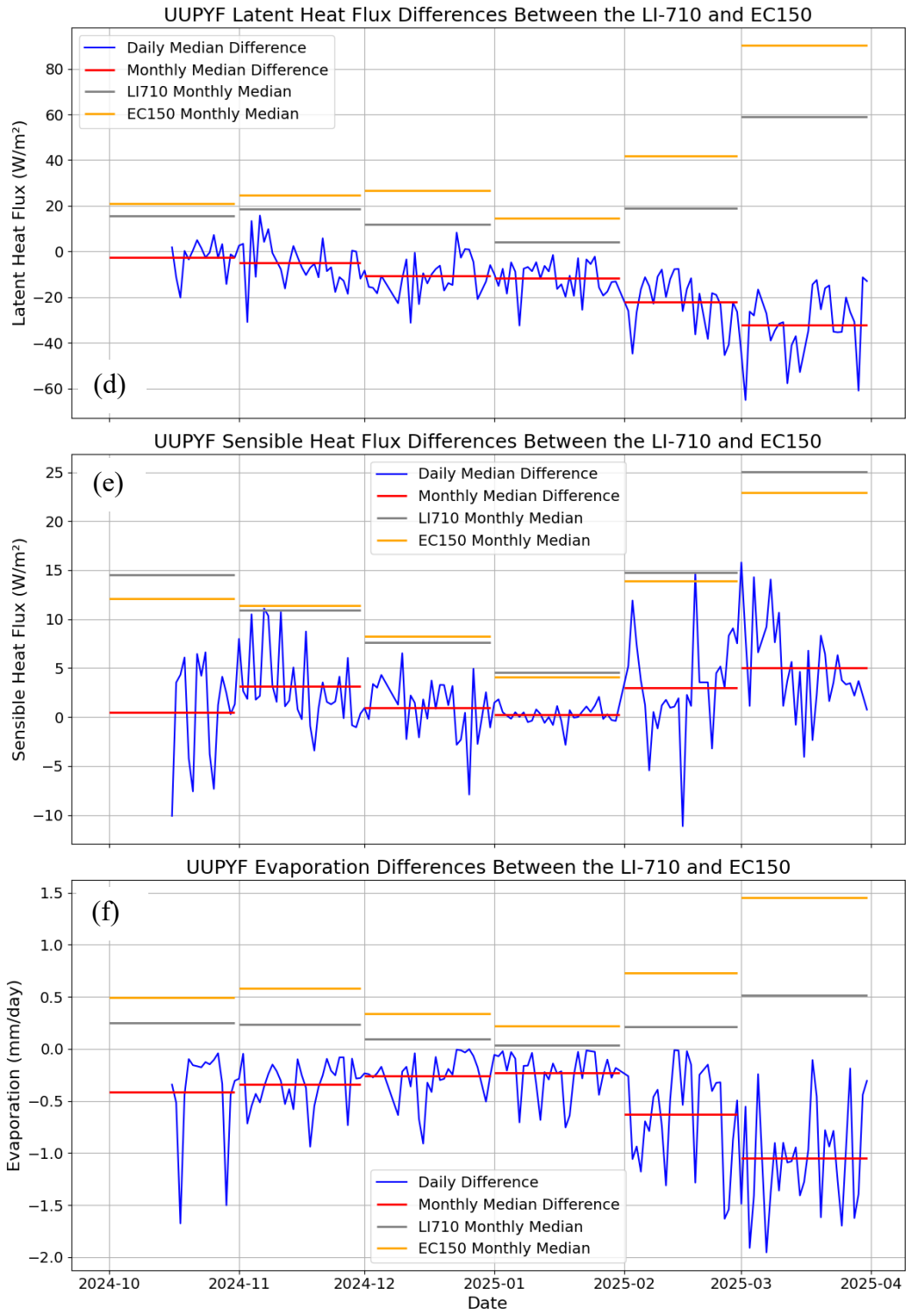


Figure 27 continued.

CHAPTER 4

DISCUSSION

This pilot study investigated momentum, sensible, and latent heat fluxes over two surface types (mineral ponds and exposed playa) within the GSL's meander line using two different eddy covariance sensor systems (Campbell Scientific EC150 and the LI-COR LI-710). The primary goal was to assess sensor performance and evaluate the feasibility of using the less expensive LI-710 as a substitute for the EC150 for long-term monitoring of evaporative fluxes in the harsh environments within the GSL Basin.

Sensors were deployed at two sites: UUCMF, located between mineral ponds operated by Compass Minerals, and UUPYF, located on the Farmington Bay playa. Data collection beginning during August 2024 is ongoing, with the focus of this study on the "cool season" defined as 16 October 2024 – 31 March 2025, with analyses focusing on *H*, LE, and evaporation rates. Key findings from this study and potential future work are discussed below.

4.1 Key Findings

4.1.1 Diagnostic Code Utility

The diagnostic flags from both sensor systems proved very useful for evaluating potential sensor performance issues. The EC150's QC flag system, based on turbulence

characteristics and stationarity tests (Foken et al., 2004, 2012), identified 30-minute samples of questionable flux measurements, particularly during weak or intermittent turbulence common during stable periods during the cool season, particularly at the UUPYF site. Of less concern due to appropriate sensor siting is the relatively infrequent periods (<10%) when the wind direction was towards or away from the 3-D sensors' booms. The LI-710's diagnostic codes also flagged performance issues, particularly when the internal cell temperature dropped below 5°C or when airflow was obstructed.

4.1.2 Environmental Conditions and Site Differences

Data quality and sensor performance varied notably between the two sites due to differences in surface characteristics and atmospheric conditions. At UUPYF, the playa surface is a frequently changing environment that undergoes substantial seasonal transitions: it can be flooded, frozen, or snow-covered during winter, flooded in spring, and typically dries out during summer and fall (Hang et al., 2015; Massey et al., 2017). This seasonal variability results in larger diurnal temperature swings and more frequent periods of boundary layer stability, particularly during calm, cold nights in winter when turbulence diminishes. These stable conditions complicate flux measurements, as eddy covariance techniques rely on turbulent mixing to accurately estimate vertical fluxes (Foken et al., 2012a). Very stable conditions ($\zeta \geq 0.4$) occurred during 9% of the cool season at UUPYF, more than double that observed at UUCMF. Likewise, very unstable conditions ($\zeta \leq -0.4$) were more common at UUPYF (16%) than UUCMF (10%), reflecting the playa's wider diurnal range and flux responses to variable surface conditions.

In contrast, the UUCMF site, located between shallow mineral ponds, maintains more consistent surface moisture levels and generally experiences less temperature variability. The presence of water in the ponds helps moderate near-surface temperatures, reducing diurnal extremes and maintaining a more neutral or unstable boundary layer state conducive to eddy covariance measurements. This more favorable environment is reflected in consistently higher momentum flux and friction velocities at UUCMF, where u* exceeded 0.1 m s^{-1} during a greater portion of the day, allowing for more valid flux measurements. In contrast, the moisture content of the playa surface at UUPYF can vary widely depending on flooding or drying cycles, further complicating flux footprint consistency (Jensen et al., 2016; Massey et al., 2017; Craft and Horel, 2019).

These environmental differences influenced not only the amount of valid data collected but also the reliability of evaporation flux estimates. During the winter months, freezing fog and frost accumulation at UUPYF were observed to interfere with sensor operation, contributing to data gaps. The LI-710 performance, which is particularly sensitive to cold conditions, was overcome largely by adding insulation (pipe wrap) to maintain internal cell temperatures above freezing as much as possible. Despite these efforts, harsh playa conditions continued to pose challenges for data collection reliability. At UUCMF, while occasional disruptions from dust, insects, and maintenance issues occurred, overall environmental conditions were more favorable for continuous data collection.

Sentinel-2 NDWI and NDMI satellite imagery emphasized the variability of the surface state at both sites, including episodic flooding at UUPYF. The imagery showed longer periods of continuous water coverage at UUCMF, with the only important

difference being the north pond, which was dry during the fall, while the playa surface at UUPYF exhibited more spatial variability. This wide range of environmental variability at the playa site throughout the duration of this project has revealed the challenges of monitoring evaporation in heterogeneous landscapes and emphasizes the need for sensor systems that can operate reliably across a broad spectrum of meteorological conditions.

4.1.3 Sensor Performance Discrepancies

Consultation with LI-COR staff provided a possible explanation for the observed underestimation of LE by the LI-710 relative to the EC150. Such underestimates have also been reported by other researchers comparing the LI-710 to fully three-dimensional eddy covariance systems (Peddinti and Kisekka, 2024). That the LI-710's one-dimensional estimates of *H* generally agree well with the EC150's three-dimensional measurements points toward an issue specific to the water vapor pathway.

Potential causes include performance limitations in the fast-response relative humidity sensor or issues with the hose and pump system supplying air to the sensor. One hypothesis proposed by LI-COR may help explain the increasing underestimates observed over the cool season (evident in Figs. 24 and 27) could be due to partial clogging in the air inlet. However, LI-COR's onboard diagnostics have not indicated substantial drops in airflow that would completely explain the trend. Contributing factors for the discrepancy between the LI-710 and EC150 estimates of latent heat flux over the course of the cool season may be long-term sensor degradation or calibration drift in the LI-710's humidity sensor. This temporal pattern is consistent with performance changes

that might arise from prolonged field deployment, especially under harsh environmental conditions.

In response to these issues, LI-COR has indicated plans to provide updated LI-710 units with a firmware upgrade intended to mitigate these problems. Evaluating whether the firmware resolves the underestimation of latent heat flux will be a key objective for the next phase of the pilot study. At this time, however, it is not possible to make a definitive recommendation on whether the LI-710 is an appropriate sensor for accurately estimating evaporation in this environment.

It is also important to note that although the LI-710 does not measure horizontal wind components and therefore cannot support footprint modeling or many standard flux corrections (Foken et al., 2004, 2012b; Rannik et al., 2012), this limitation does not necessarily preclude its application near the Great Salt Lake. In this setting, sensors can be deployed in areas with relatively uniform land cover and surface characteristics (as has been done with UUCMF and UUPYF), where flux footprints are less variable and source area assumptions are more stable. As a result, despite its limitations, the LI-710 may still be useful to deploy in combination with basic meteorological sensors to measure horizontal wind and net solar and infrared radiation.

4.1.4 Estimating Evaporation

While the focus of the pilot study is to provide recommendations on ways to measure evaporation near the Great Salt Lake, preliminary daily evaporation estimates can be provided from the EC-150 sensors for the cool season. Current evaporation estimates derived from this study, conducted during the cool season are 0.52 mm day^{-1}

for playa and 0.53 mm day^{-1} for mineral ponds. Since these values are obtained after excluding substantive numbers of samples when evaporation may be small but cumulatively non-negligible, actual estimates may be higher. It is anticipated that less data will be excluded during the warm season (April-September) such that the estimates for the mineral pond and playa sites will be potentially more accurate for comparison to earlier studies. Prior estimates show a yearly average of 0.45-0.7 mm day^{-1} from the playa (Malek, 2003; Hang et al., 2015) and 3-5mm day^{-1} from mineral ponds (Turk, 1970).

4.2 Future Work

This pilot study has laid the groundwork for evaluating lower-cost eddy covariance sensor alternatives in the GSL basin, but many questions remain – particularly regarding the representativeness, reliability, and usability of evaporation estimates across all surface conditions and seasons. This research contributes to the broader GSL Basin Integrated Plan (Bingham et al., 2023) to help improve future monitoring strategies that inform water resource management policies in the face of ongoing lake desiccation and water scarcity challenges (Wurtsbaugh et al., 2017; Perry, 2024).

Future efforts should prioritize a more holistic approach to assessing sensor performance. While the EC150 is well established and treated here as truth, it is also expensive and requires substantial maintenance. Crucially, EC150 data were excluded more than 50% of the time at both sites during the cool season due to turbulence-based QC filtering. For many research applications, discarding such a large proportion of data is not desirable.

To determine whether such stringent filtering is necessary, future work should begin by evaluating the performance of the LI-710 during warm-season conditions when turbulence is expected to be stronger and more consistent. Specifically:

1. Establish warm-season reliability of the LI-710 during periods when the EC150 indicates favorable flux conditions. If the LI-710 aligns well with the EC150 during these periods, confidence in its performance under optimal conditions will increase.
2. Quantify the amount of evaporation that occurs during periods when the EC150 indicates unreliable conditions. Rather than discarding these intervals outright, future work should explore how much is being excluded and whether it meaningfully alters daily or seasonal totals.
3. Examine what the LI-710 reports during these same low-turbulence or stable periods. If it produces reasonable flux estimates despite high EC150 QC flags, this could suggest that it remains useful in operational settings.
4. Evaluate whether basic meteorological observations can help infer expected flux quality**.** If the LI-710 is deployed with inexpensive meteorological sensors (e.g., temperature, wind, humidity radiation) can simple thresholds (e.g., friction velocity, wind direction) predict when the EC150 would have flagged the flux as unreliable?
5. Reconsider the assumption that stable conditions always imply bad fluxes. While turbulence is suppressed, the resulting shift in footprint may not be problematic if the underlying surface is spatially uniform. For the GSL playa and mineral pond

environments, this assumption may hold, reducing the need for aggressive filtering.

6. Compare evaporation estimates from these ground-based sensors to those from satellite and modeling products. Unlike eddy covariance systems, satellite retrievals and models do not discard stable or low-turbulence periods. Thus, heavily filtered sensor data may not align with estimates from coarser-resolution sources, which often integrate over all times and conditions.

An important extension of this work is to develop more robust approaches for estimating uncertainty. Rather than assuming that QC-filtered fluxes represent ground truth, future efforts should consider including all data – regardless of QC flags – and evaluate how total evaporation changes. This difference can be interpreted as an uncertainty range. Such an approach should be applied to both EC150 and LI-710 datasets to understand how sensitive evaporation estimates are to QC assumptions.

Finally, collecting salinity data from the mineral ponds at UUCMF should be a priority. Given the strong influence of salinity on vapor pressure and evaporation rates, having access to this variability would enable better comparison with model-based estimates and could inform flux corrections under different salinity conditions.

A firmware upgrade from LI-COR in summer 2025 will be evaluated through the remainder of the pilot study to assess whether the LI-710 underestimate of LE and evaporation rate is overcome. Additionally, expanding monitoring to other land cover types within the GSL Basin, including open water areas will be critical for basin-wide evaporation estimates (Lobos-Roco et al., 2022; Gu et al., 2024).

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